



Image Processing

Natsuda Kaothanthong, Ph.D.

Sirindhorn International Institute of Technology
Thammasat University



Natsuda Kaothanthong

Associate Professor



Education

- **Ph.D., Information Sciences**
 - Graduate School of Information Sciences, Tohoku University
- **M.S., Information Sciences**
 - Graduate School of Information Sciences, Tohoku University



Work Experiences

- **Research Fellow**
 - Japan Society for the Promotion of Science (JSPS)
- **Assistant Professor**
 - School of Management Technology, SIIT
- **Associate Professor**
 - School of Management Technology, SIIT
- **Committee**
 - Artificial Intelligence Association of Thailand (AIAT)



Research

- **Pattern Recognition and Image Processing**
 - Research collaboration on *Customer's Dunning Action Prediction*
 - *ATM fraud* detection
 - A model for *generating ICGA image* from colored fundus

- **Machine Learning and Artificial Intelligence**
 - A deep learning model for *classifying colored fundus* images
 - A deep learning model for *classifying diagnosis of anti-VEGF* treatments using OCT images
 - *Locating thrombus* for Acute Ischemic Stroke patient using ncCT
 - *Locating Brain Infarct* using ncCT



Contact

natsuda@siit.tu.ac.th

Agenda



01. Imaging
Fundamental
Image Acquisition

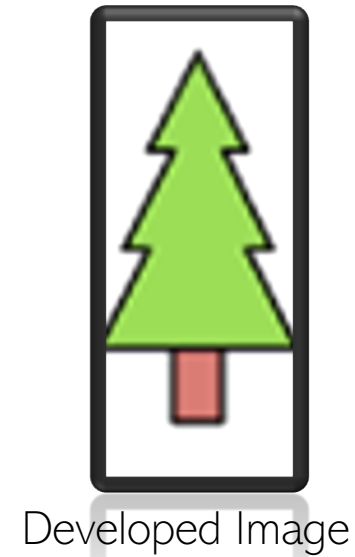
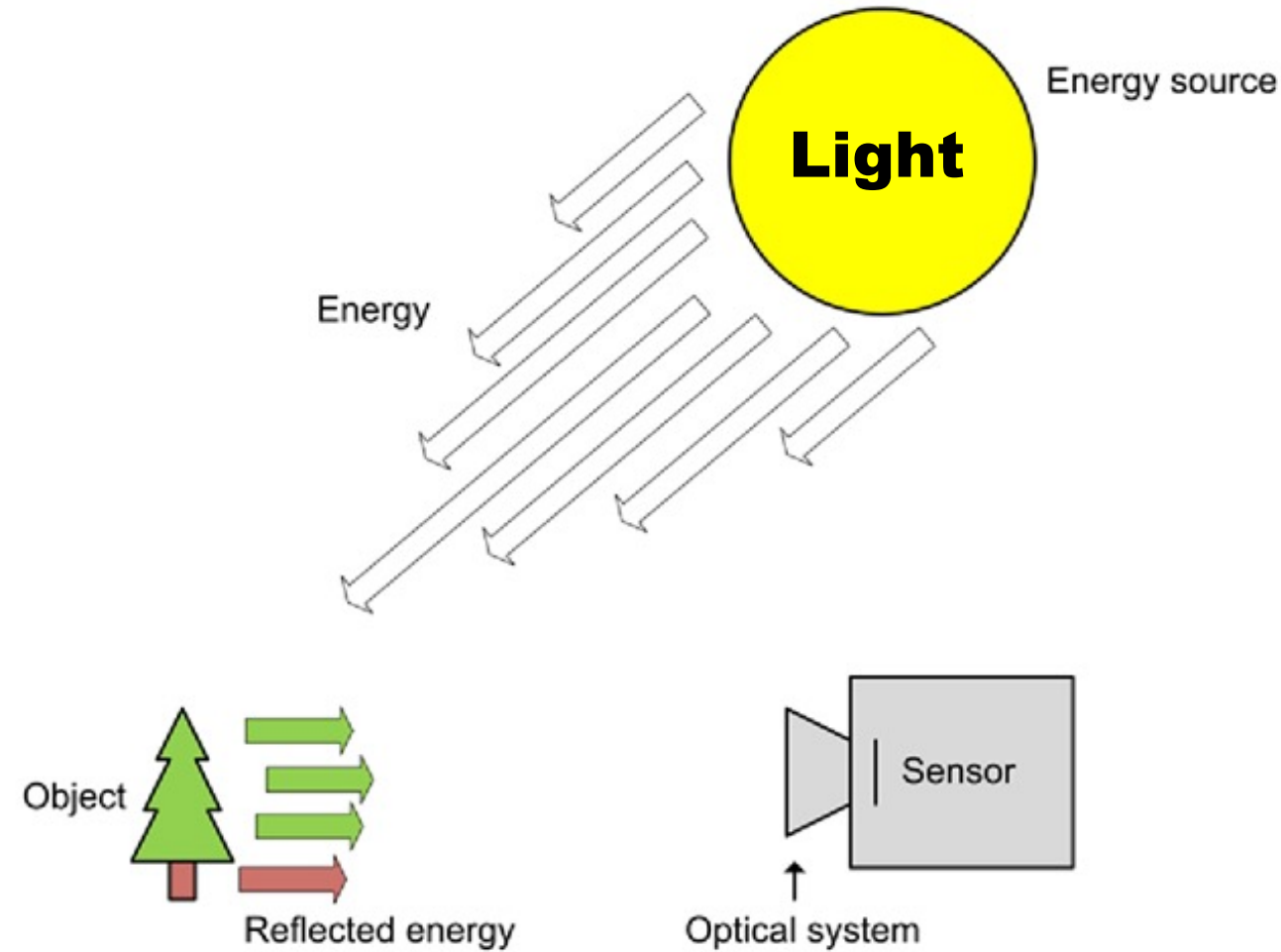
02. Medical Imaging
Types

03. Fundamental of
Image Processing
Basic processing
Spatial Filter

04. Morphological
Technique
Morphology

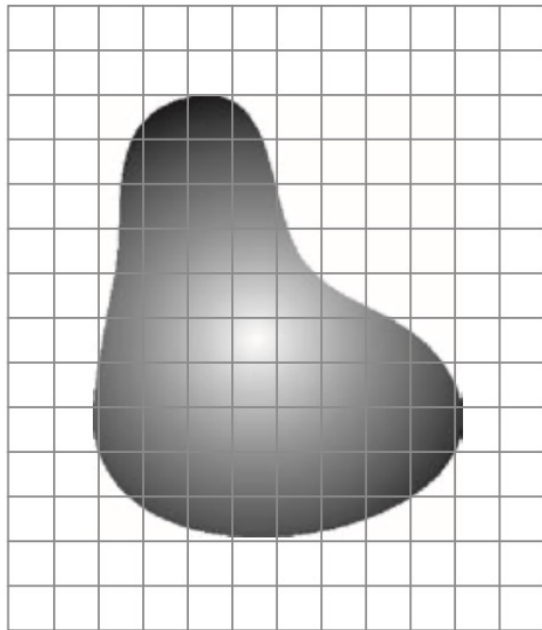
05. Ultrasound

Image Acquisition

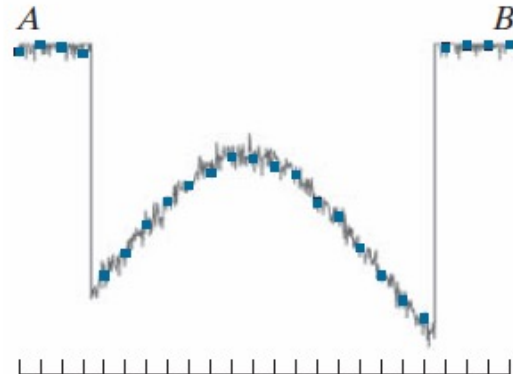


187	180	170	162	130	82	38	22	13	16
179	184	182	175	155	109	53	19	19	18
180	183	182	179	165	125	68	29	22	19
181	181	181	182	176	145	90	45	26	20
183	181	181	184	183	160	110	60	29	21
187	184	184	185	185	169	125	73	33	22
186	186	185	184	182	172	135	83	35	22
186	188	187	182	179	174	142	92	38	22
188	190	190	183	179	176	146	97	40	23
188	191	187	187	182	178	152	107	53	24

Acquiring Digital Image



Projected real object



Signal from the sensor

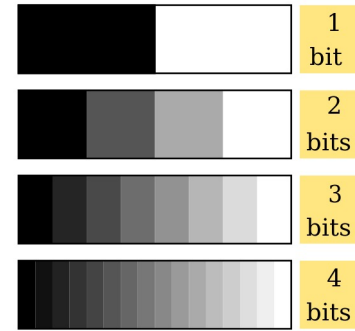
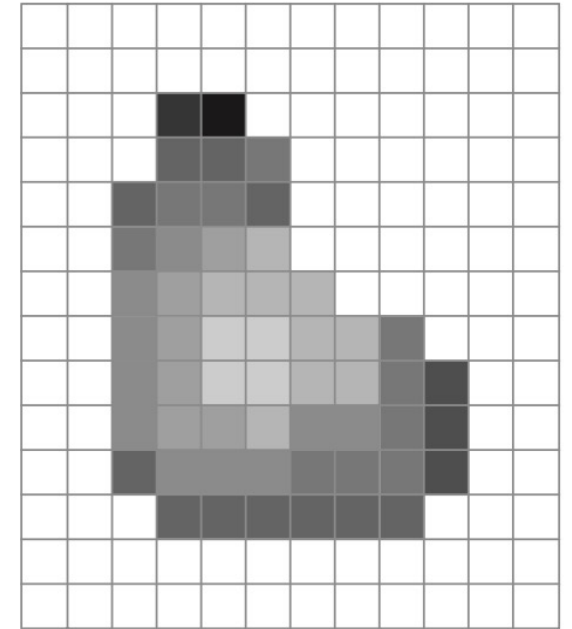
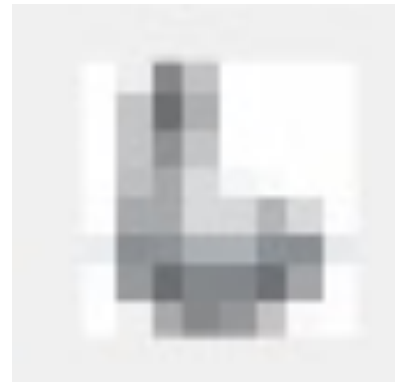


Image Quantization



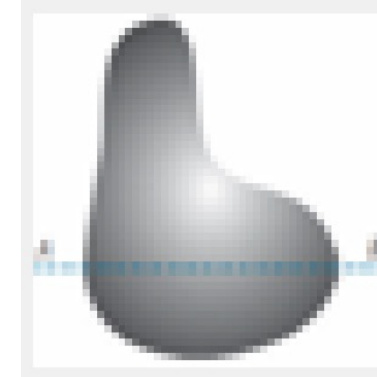
Digital Image



8×8 grid size



20×20 grid size



50×50 grid size

Image Sampling

Energy Source

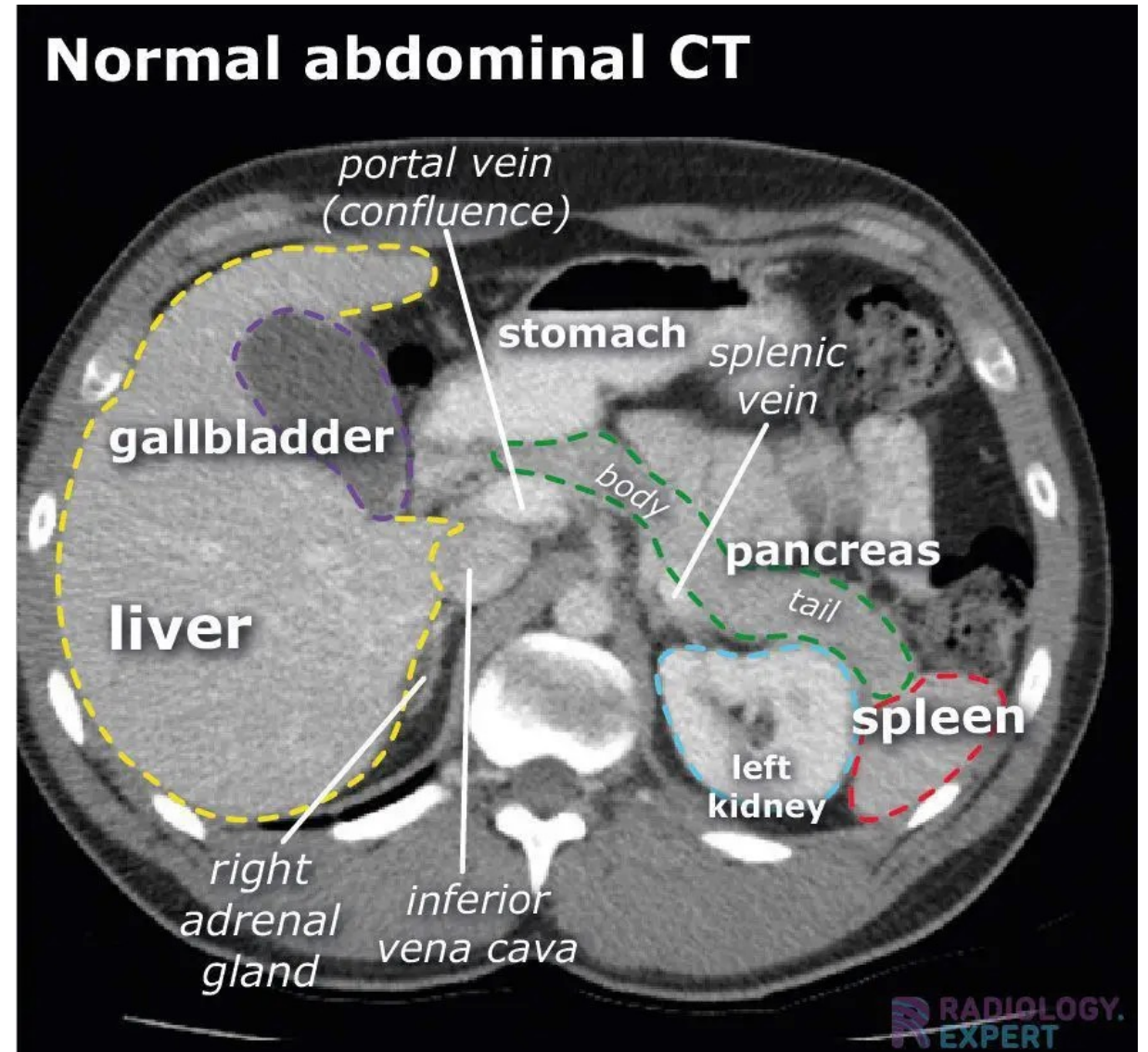
Signal Digitization



Medical Image

Purpose

To visualize interior of a body for clinical analysis and medical intervention, as well as visual representation of the **function** of some organs or tissues.



Medical Imaging

Energy Source

X-rays: Ionizing radiation

- Radiography
- Computed Tomography (CT) Scan

Sound waves

- Ultrasound

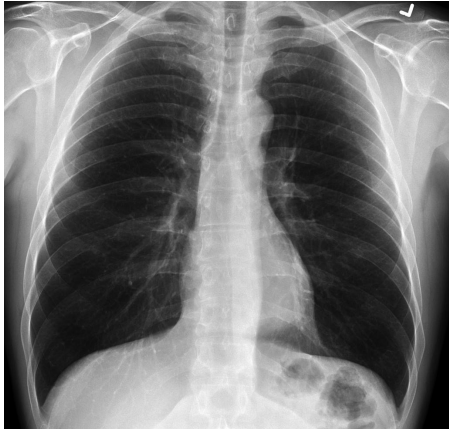
Magnetic fields/radiofrequency waves

- Magnetic Resonance Imaging (MRI)

Gamma rays

- Nuclear Medicine

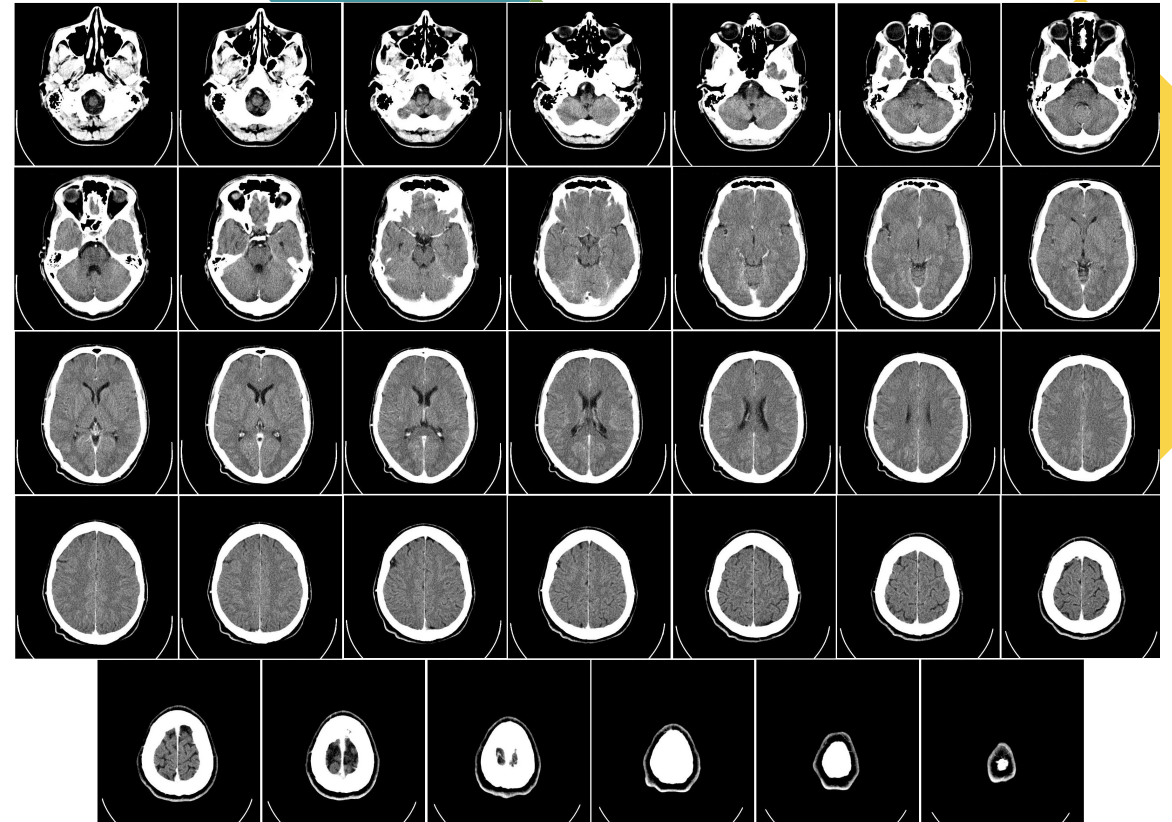
X-ray



Chest X-ray



Projection Radiography

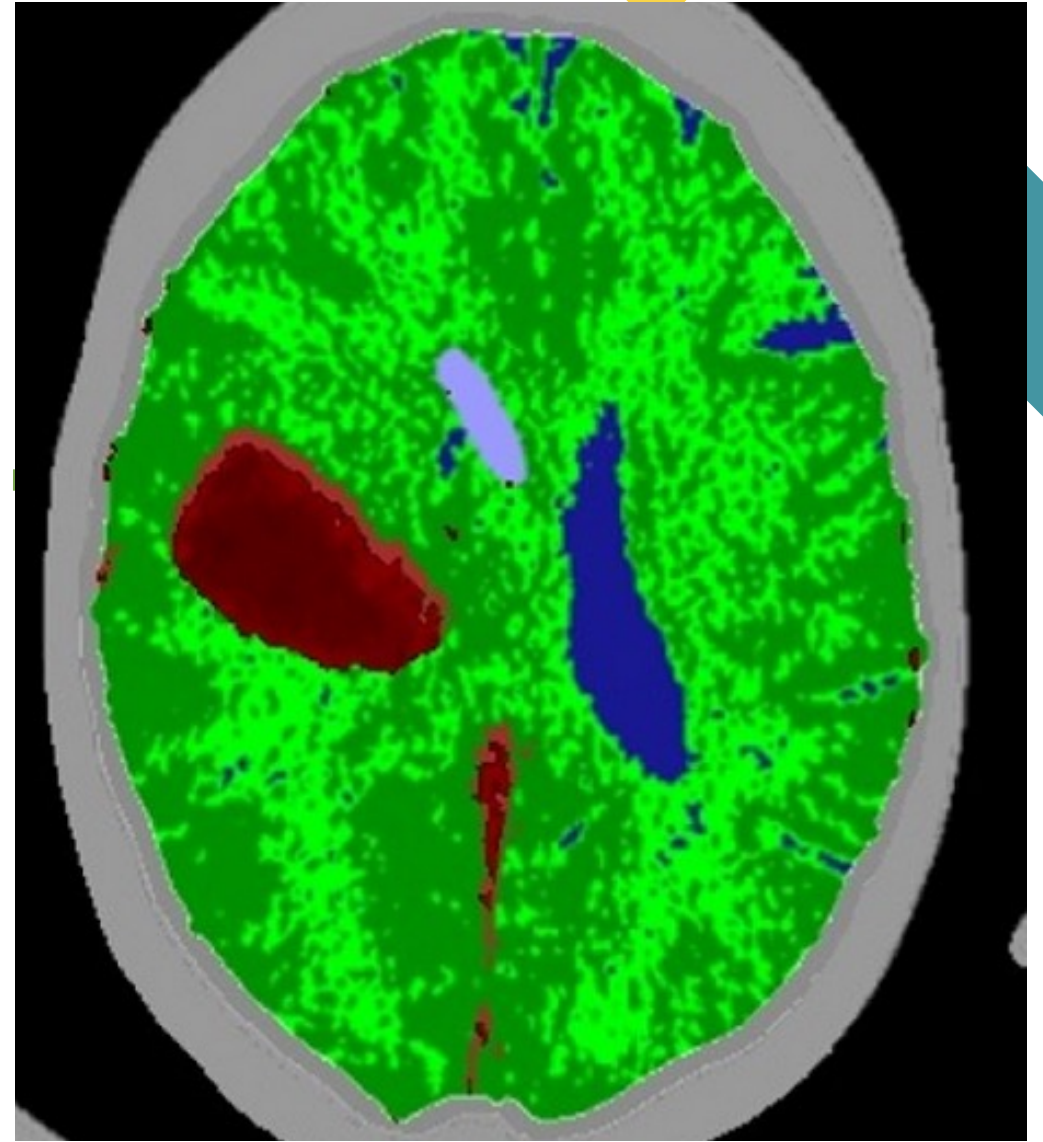
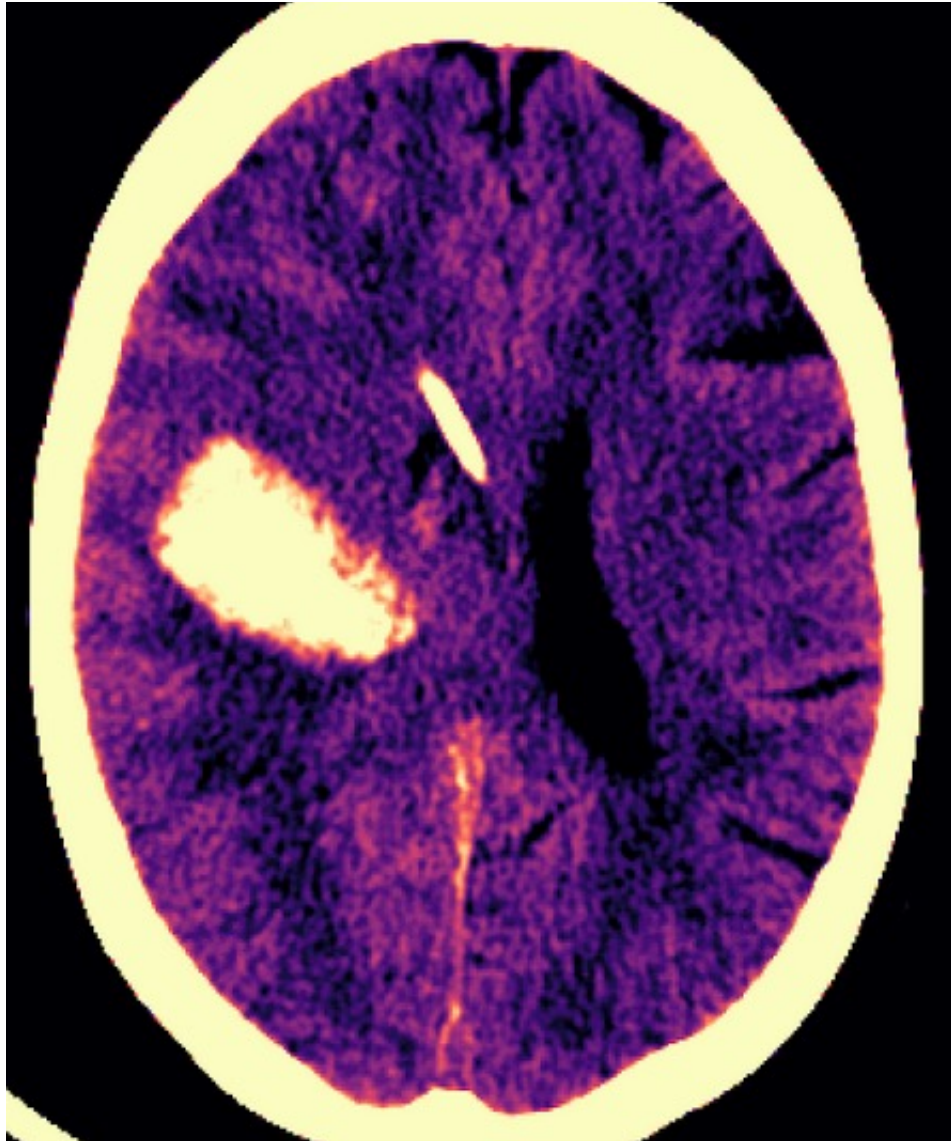


Brain CT

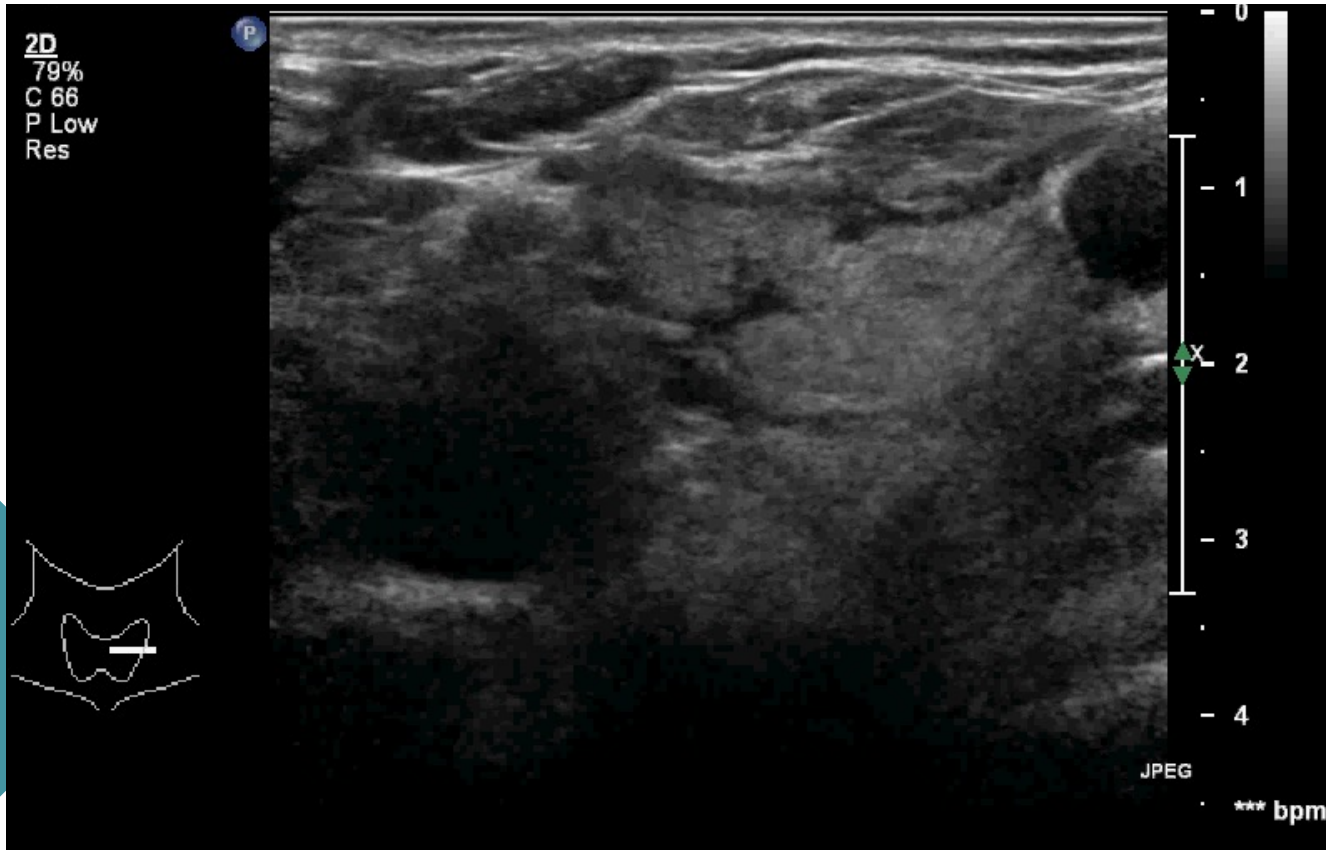
Computed Tomography

Sources:

- <https://radiopaedia.org/cases/normal-chest-x-ray>
- https://en.wikipedia.org/wiki/Computed_tomography_of_the_head
- Wikipedia



Ultrasound Image



Thyroid Nodule

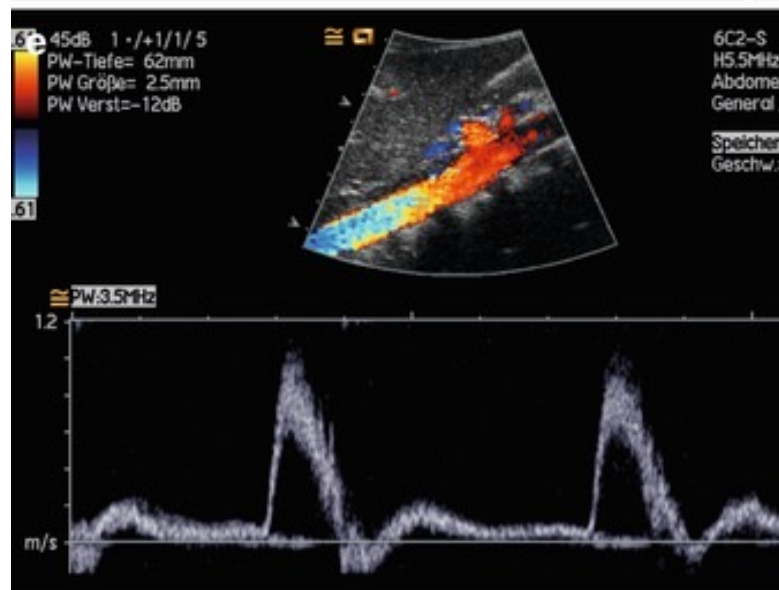
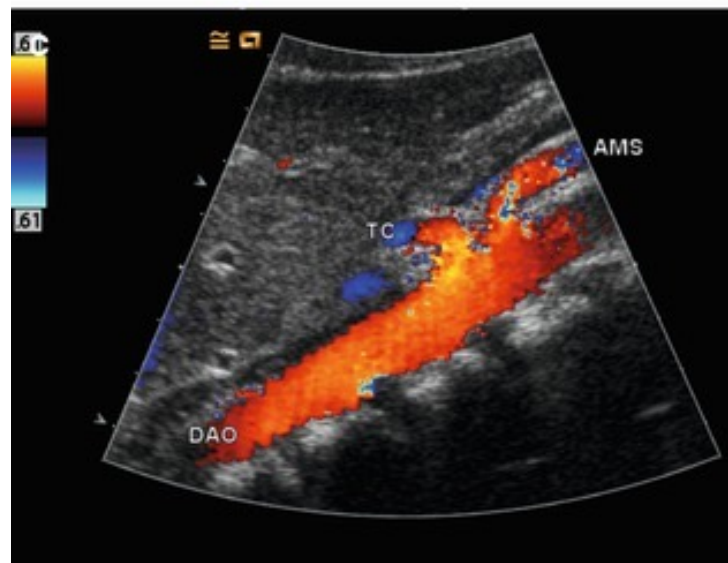


Anterior Segment (AS) of the eye

Source:

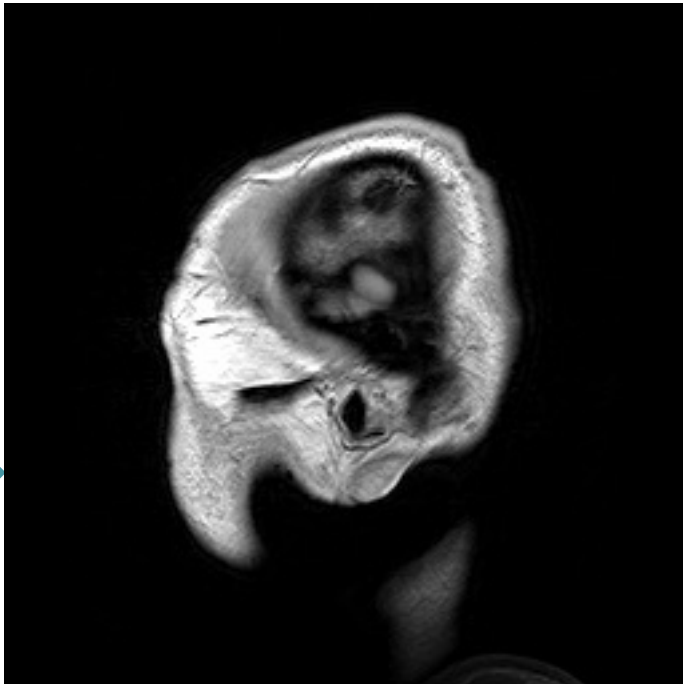
- <https://drrahmanultrasounds.com/thyroid-nodule/>
- Wikipedia

Doppler



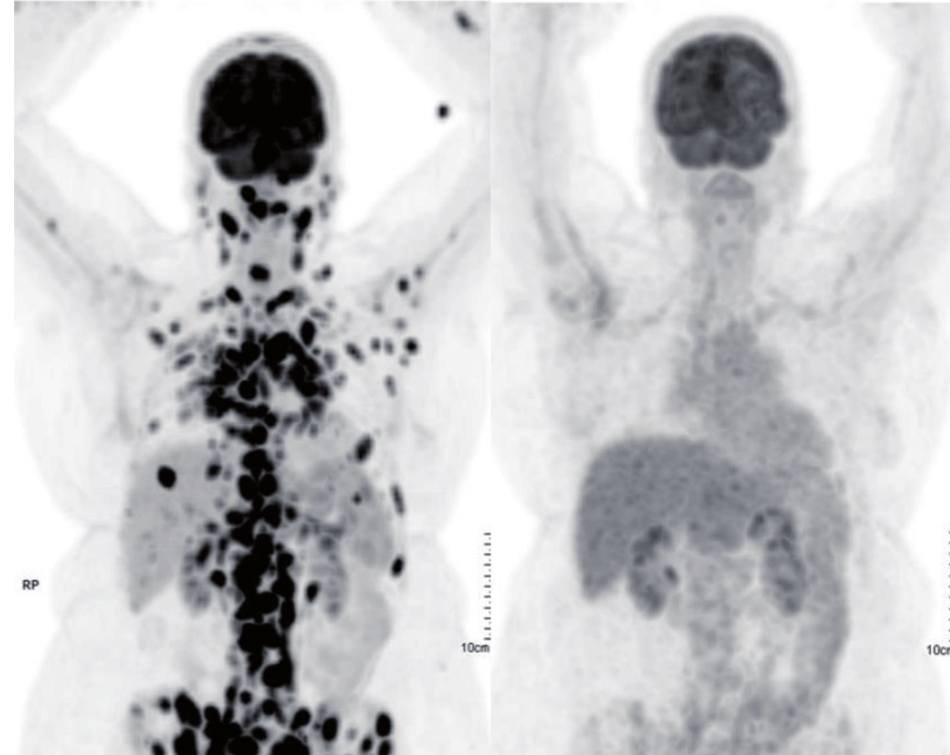
MRI and PET

MRI



- https://www.researchgate.net/figure/PET-CT-scan-showing-marked-improvement-after-initiation-of-immunosuppressive-therapy-with_fig1_354383100
- <https://www.regencymedicalcentre.com/blog/ct-scan-vs-mri>

PET CT

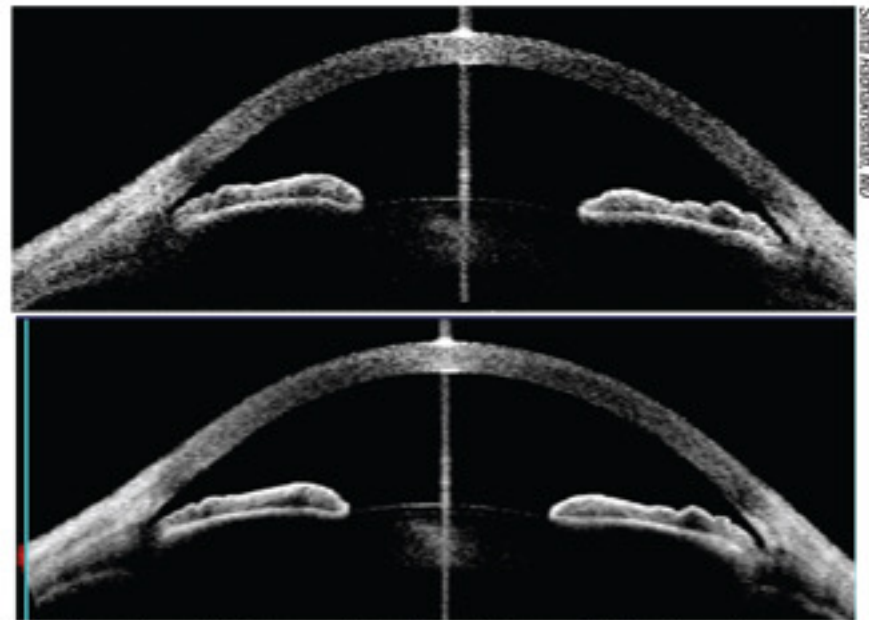


Other Types

Fundus Image

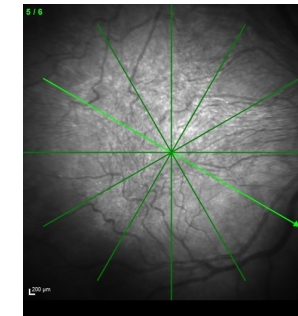


Optical coherence tomography (OCT)

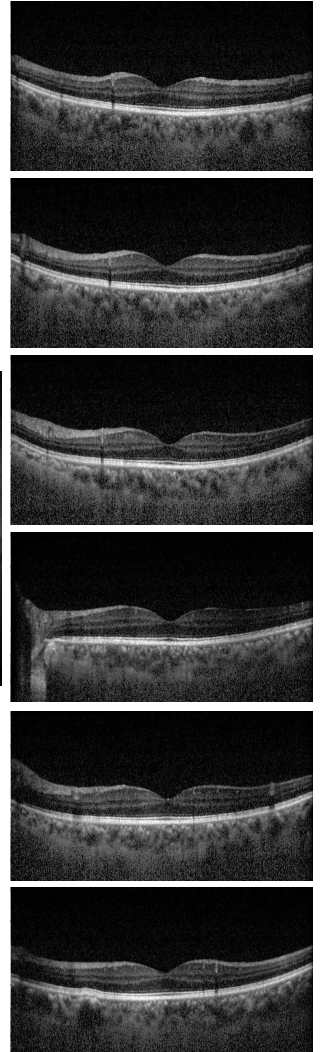


Using the enhanced anterior segment single scan feature on the Visante reveals more detail because it averages four scans, improving the signal-to-noise ratio (bottom image).
Top image: Standard anterior segment single scan.

- AS-OCT from 4 quadrants



- OCT of 6 axis



Format

DICOM: Digital Imaging and Communications in Medicine

- Used for storing and transmitting medical images devices
- It groups information into data sets
 - Image, Patient ID, Patient information, etc.
 - (0028,1050) DS Window Center
 - (0028,1051) DS Window Width
 - <https://www.dicomlibrary.com/dicom/dicom-tags/>

Search for DICOM Tags:

0002,0000	US	patient
Tag	VR	Name
(0002,0000)	UL	File Meta Information Group Length
(0002,0001)	OB	File Meta Information Version
(0002,0002)	UI	Media Storage SOP Class UID
(0002,0003)	UI	Media Storage SOP Instance UID
(0002,0010)	UI	Transfer Syntax UID
(0002,0012)	UI	Implementation Class UID
(0002,0013)	SH	Implementation Version Name
(0002,0016)	AE	Source Application Entity Title
(0002,0017)	AE	Sending Application Entity Title
(0002,0018)	AE	Receiving Application Entity Title
(0002,0100)	UI	Private Information Creator UID
(0002,0102)	OB	Private Information
(0004,1130)	CS	File-set ID
(0004,1141)	CS	File-set Descriptor File ID
(0004,1142)	CS	Specific Character Set of File-set Descriptor File
(0004,1200)	UL	Offset of the First Directory Record of the Root Directory Entity
(0004,1202)	UL	Offset of the Last Directory Record of the Root Directory Entity
(0004,1212)	US	File-set Consistency Flag
(0004,1220)	SQ	Directory Record Sequence
(0004,1400)	UL	Offset of the Next Directory Record
(0004,1410)	US	Record In-use Flag
(0004,1420)	UL	Offset of Referenced Lower-Level Directory Entity
(0004,1430)	CS	Directory Record Type
(0004,1432)	UI	Private Record UID
(0004,1500)	CS	Referenced File ID
(0004,1504)	UL	MRDR Directory Record Offset
(0004,1510)	UI	Referenced SOP Class UID in File
(0004,1511)	UI	Referenced SOP Instance UID in File
(0004,1512)	UI	Referenced Transfer Syntax UID in File
(0004,151A)	UI	Referenced Related General SOP Class UID in File
(0004,1600)	UL	Number of References
(0008,0001)	UL	Length to End
(0008,0005)	CS	Specific Character Set
(0008,0006)	SQ	Language Code Sequence
(0008,0008)	CS	Image Type

Agenda

01. Imaging Fundamental

Image Acquisition

02. Medical Imaging

Types

03. Fundamental of Image Processing

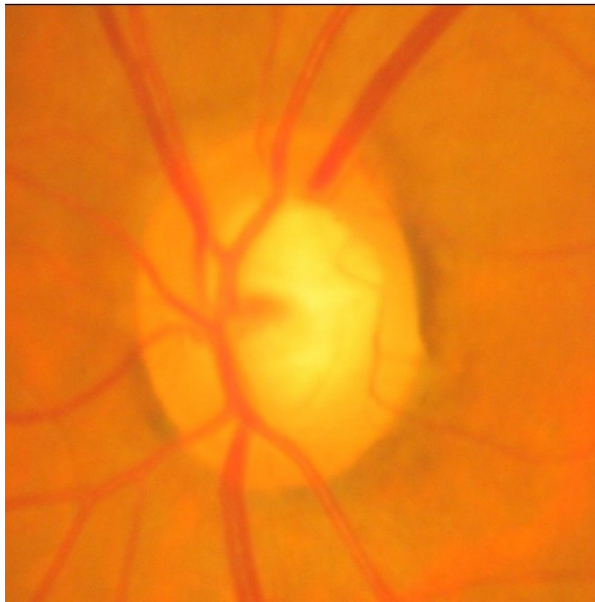
Basic processing
Spatial Filter



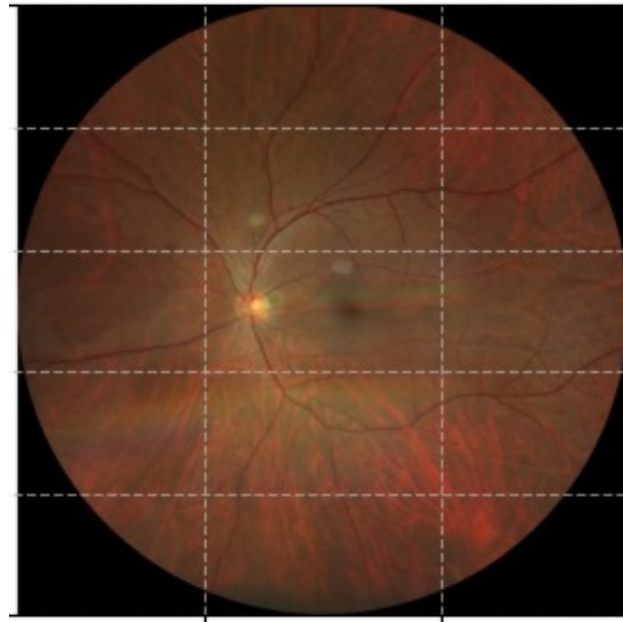
Original Image



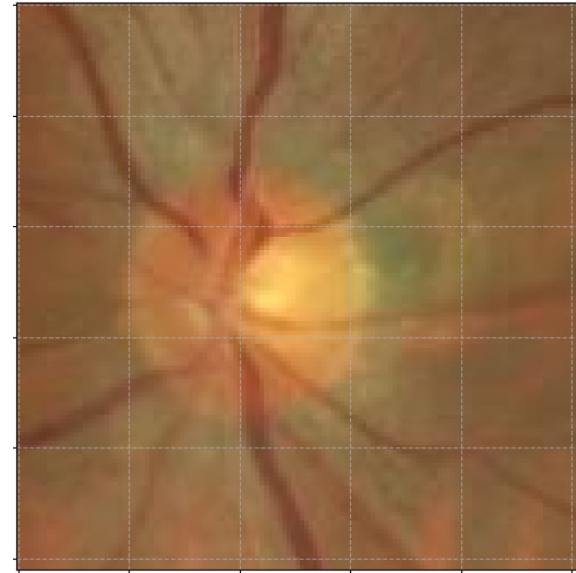
Optic Disc Cropped (Narrower)



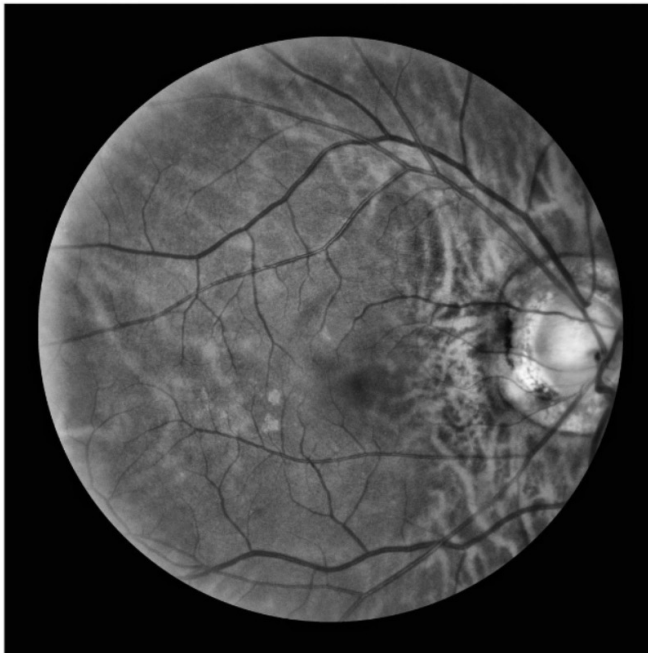
Original Image



Optic Disc Cropped (Narrower)



Enhanced Image (CLAHE)



Binary Thresholding

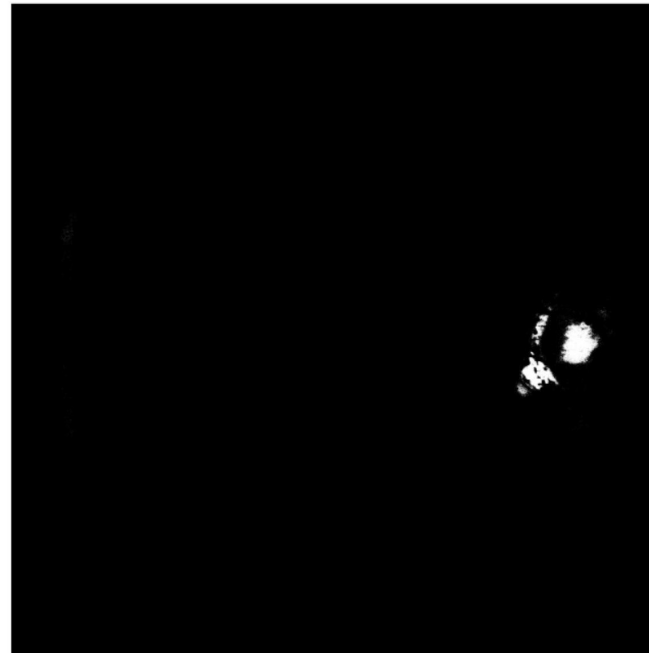


IMAGE ENHANCEMENT

Intensity Transformation

Histogram Processing

Spatial Filtering

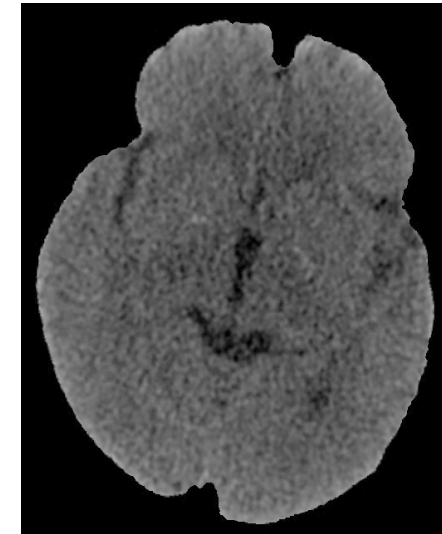
Smoothing

Sharpening

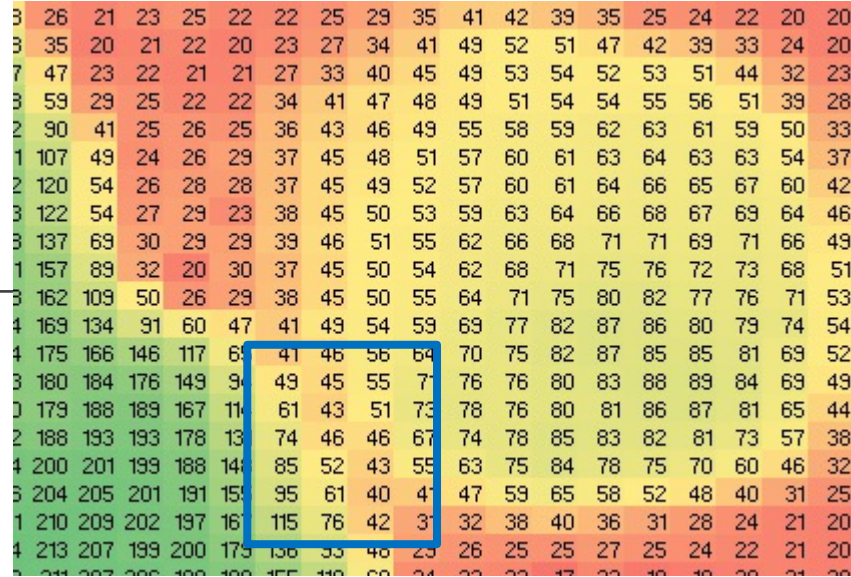
03 FUNDAMENTAL OF IMAGE PROCESSING

Image Enhancement

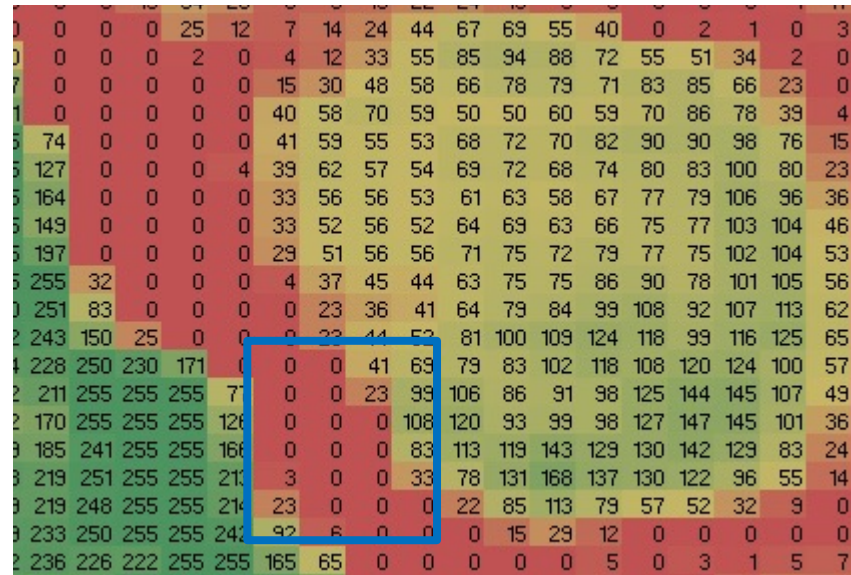
- Enhancement:
 - A process of manipulating an image so that the result is more suitable than the original for a specific application.
 - Application: Image segmentation
 - Enhancement techniques to be applied is a problem-oriented.
 - Natural-scene image segmentation
 - Brain CT image segmentation



Example



Area of an object and the background is **hard** to be separated.



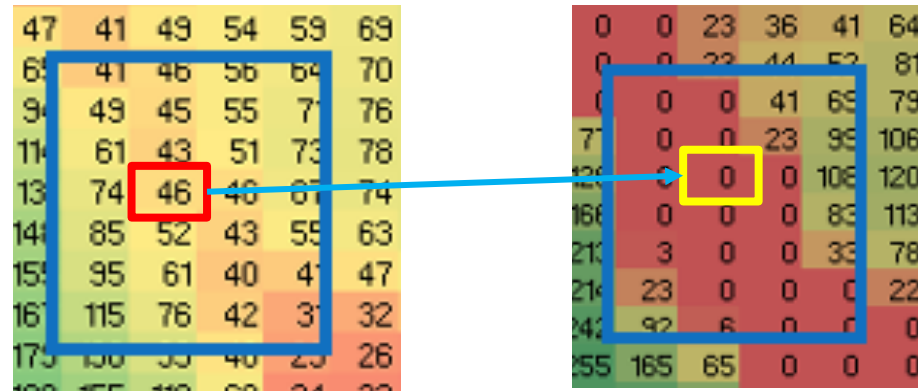
Area of an object and the background **can** be separated.

Intensity Transformation

The simplest image processing techniques.

Given a pixel value in the digital value r , the transformed value is denoted by s

- $s = T(r)$
- where T is a transformation function such as $s = \max\{0, r - 50\}$



Intensity Transformation

Image Negative:

To reverse the intensity level of a digital image.

Each pixel can be transformed using a linear function

$s = L - I - r$, where L is the highest intensity value

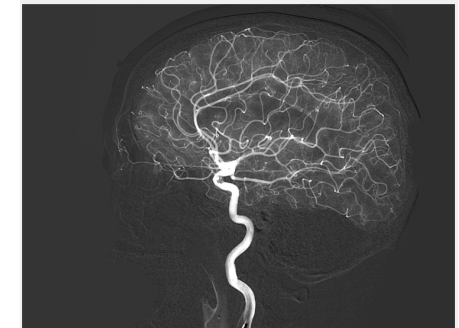
Advantage:

Enhancing gray detail in the dark regions of an image.

Enhancing the mammogram image.



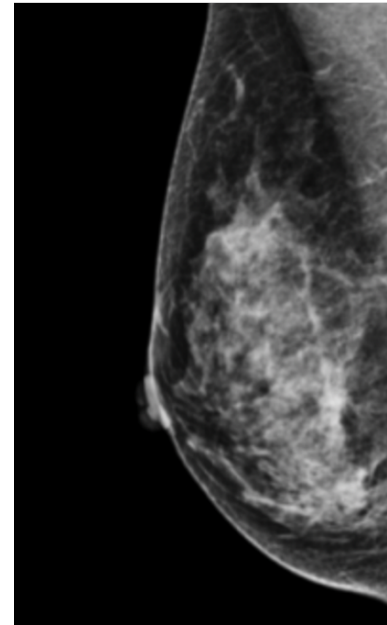
Original: Cerebral angiography image
<https://www.healthline.com/health/cerebral-angiography>



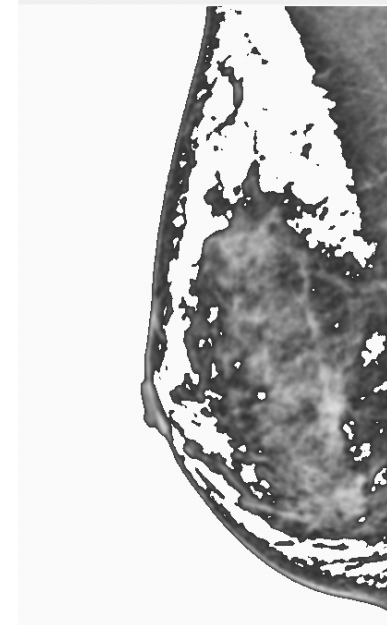
Negative Image

Intensity Transformation

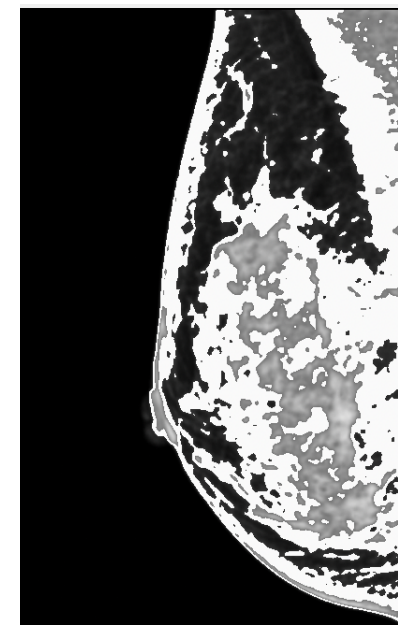
- **Intensity-Level Slicing:**
 - To highlight a specific range of intensities in an image.
 - Only the value of uninterested pixels is changed.
- Given an interested intensity range A and B , each pixel is transformed using the following condition
 - If $r \in [A, B]$ then $s = r$
 - Otherwise, $s = 255$
- **Advantage:**
 - Enhancing the interested region such as cancer to allow a size measurement.



Original Mammogram



$A = 1$ and $B = 50$



$A = 50$ and $B = 120$

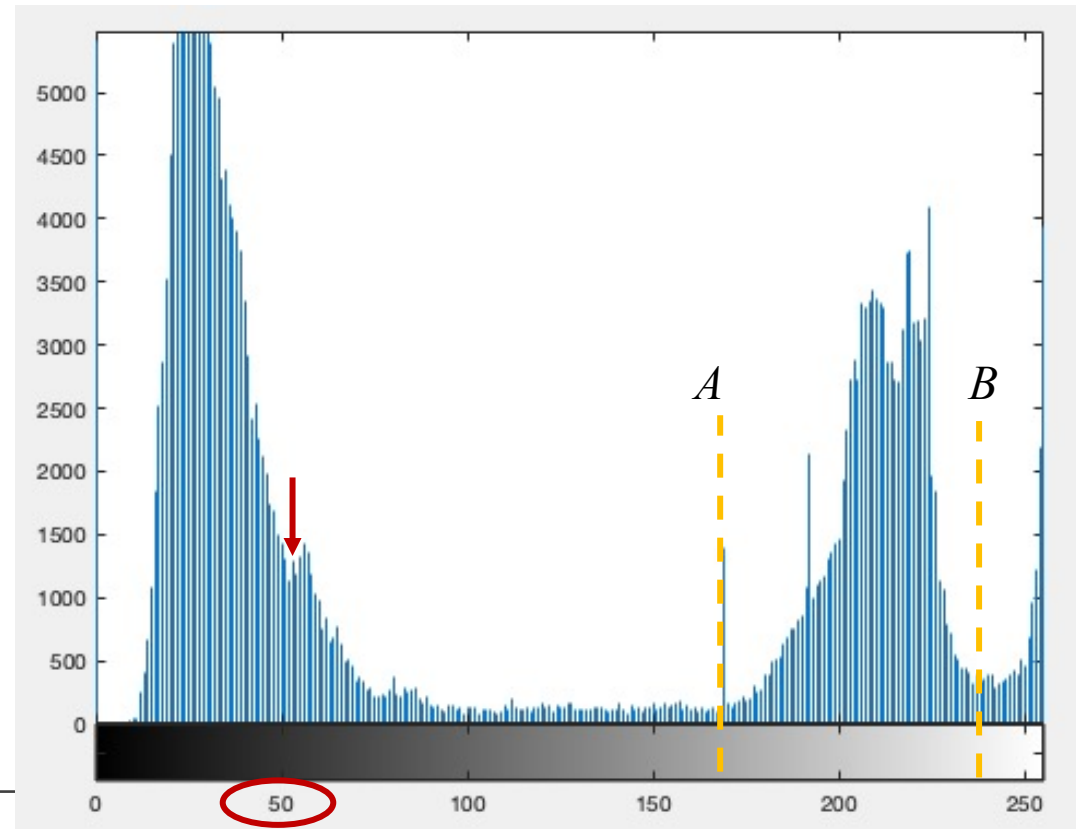
Histogram Processing

A **histogram** shows the distribution of the pixels in the image according to the intensity value of each pixel.

Let r_k , for $k = 0, \dots, L - 1$ denotes the intensity of an L -level digital image of a pixel $f(x, y)$.

The histogram of an image is obtained by counting the number of pixel $f(x, y)$ whose intensity is equal to r_k .

- The number of pixel is denoted by n_k



Histogram of 256 intensity level

Majority of pixels in the image is dark (*low intensity value*) and follow by white pixel (*high intensity value*)

Histogram Processing

24	26	29	37	45
26	28	28	37	45
27	29	23	38	45
30	29	29	39	46
32	20	30	37	45
50	26	29	38	45
91	60	47	41	49
146	117	65	41	46
176	149	94	49	45
189	167	114	61	43
193	178	131	74	46
199	188	148	85	52
201	191	155	95	61
202	197	167	115	76
199	200	179	136	93

The **histogram bins** is the number of pixels whose intensities are in the range $[a, b)$

- For example: For $L = 256$ and the desired histogram bins is 8, the interval of each bin is 32 intensity level.

Normalized histogram is the probabilities of intensity level occurring in an image.

Histogram Bins

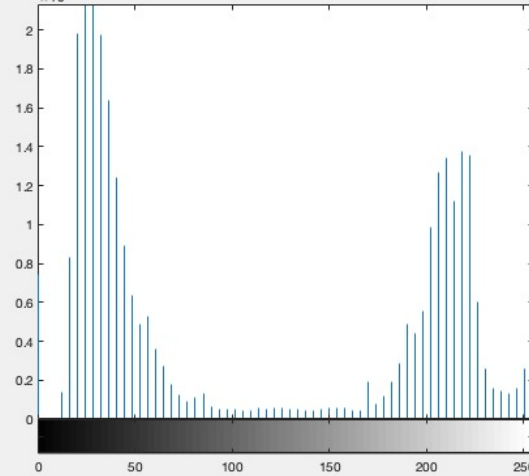
16	23	4	2	4	6	8	0
$[0, 32)$	$[32, 64)$	$[64, 96)$	$[96, 128)$	$[128, 160)$	$[160, 192)$	$[192, 224)$	$[224, 256)$

Normalized Histogram

0.2539	0.3650	0.0063	0.0317	0.0634	0.09523	0.12698	0
$[0, 32)$	$[32, 64)$	$[64, 96)$	$[96, 128)$	$[128, 160)$	$[160, 192)$	$[192, 224)$	$[224, 256)$

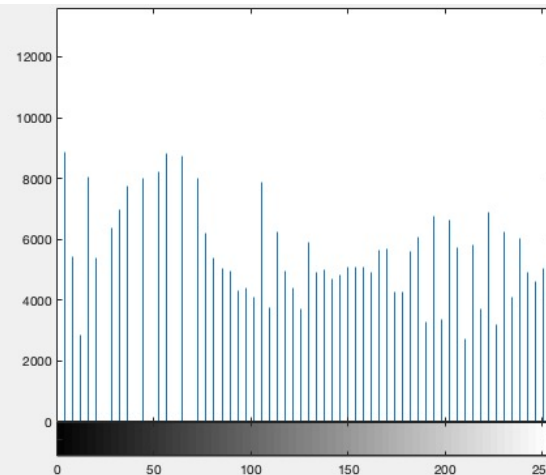
Histogram Equalization

Original



A histogram processing method to adjust the contrast of the image to have an evenly distributed intensity throughout the range.

Histogram Equalization



Local Histogram Processing

The method using a histogram processing is a **global** method.

- The transformation function is based on the intensity distribution of an entire image.

Local histogram processing aims to enhance details over small areas in a image.



Spatial Filtering

Recap - Neighborhood Operation

Let S_{xy} be a set of coordinates of a neighborhood of (x, y) in a image f

The value of the pixel (x, y) in the image g is determined by a specific operation on the neighborhood S_{xy}

For example: A mean operation

187	180	170	162	130	82	38	22	13	16
179	184	182	175	155	109	53	19	19	18
180	183	182	179	165	125	68	29	22	19
181	181	181	182	176	145	90	45	26	20
183	181	181	184	183	160	110	60	29	21
187	184	184	185	185	169	125	73	33	22
186	186	185	184	182	172	135	83	35	22
186	188	187	182	179	174	142	92	38	22
188	190	190	183	179	176	146	97	40	23
188	191	187	187	182	178	152	107	53	24

Intensity value on an image f

Intensity value on an image g

Spatial Filtering

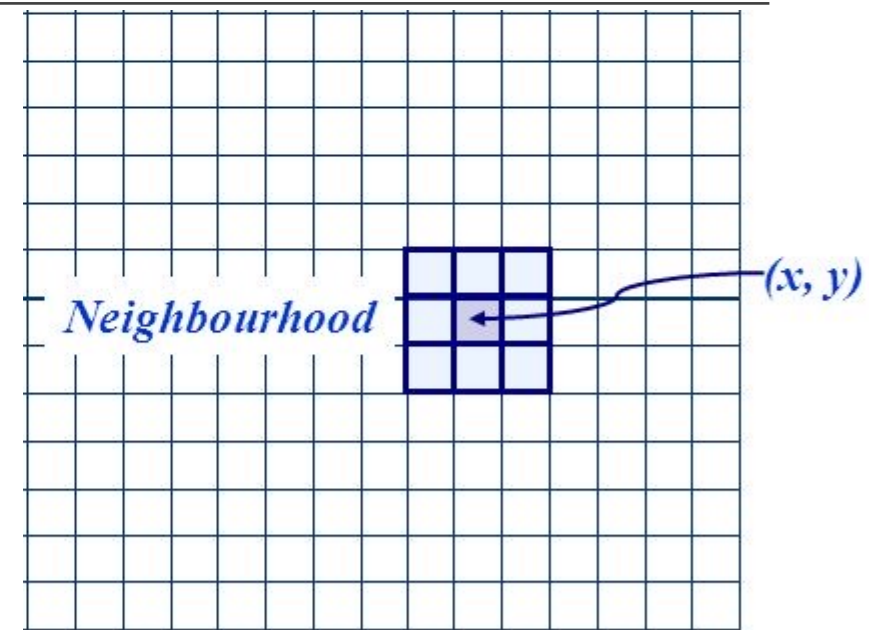
Filtering is to pass, modify, or rejecting specified frequency component of an image.

Highpass filter: A filter that passes high frequencies to sharpen an image.

Lowpass filter: A filter that passes low frequencies to smoothen an image.

Spatial filtering: Allow a modification of an image by replacing the value of each pixel by a function of the values of the pixel and its neighbors.

- Smoothing
- Sharpen



An image

Linear Spatial Filter

It performs a sum-of-products operation between an image f and a filter kernel w

- Filter kernel is also called: a mask, a template, or a window

	$f(x-1, y-1)$	$f(x, y-1)$	$f(x+1, y-1)$	
	$f(x-1, y)$	$f(x, y)$	$f(x+1, y)$	
	$f(x-1, y+1)$	$f(x, y+1)$	$f(x+1, y+1)$	

image f

$w(-1, -1)$	$w(0, -1)$	$w(1, -1)$
$w(-1, 0)$	$w(0, 0)$	$w(1, 0)$
$w(-1, 1)$	$w(0, 1)$	$w(1, 1)$

Filter kernel w

-1	0	1
-2	0	2
-1	0	1

Sobel Filter

0	-1	0
-1	4	-1
0	-1	0

Laplacian Filter

	$f(x-1,y-1)$	$f(x,y-1)$	$f(x+1,y-1)$	
	$f(x-1,y)$	$f(x,y)$	$f(x+1,y)$	
	$f(x-1,y+1)$	$f(x,y+1)$	$f(x+1,y+1)$	

image f

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

$w(-1,-1)$	$w(0,-1)$	$w(1,-1)$
$w(-1,0)$	$w(0,0)$	$w(1,0)$
$w(-1,1)$	$w(0,1)$	$w(1,1)$

filter kernel w

$$\begin{aligned} g(x, y) = & w(-1, -1)f(x - 1, y - 1) + w(0, -1)f(x, y - 1) \\ & + w(1, -1)f(x + 1, y - 1) + w(-1, 0)f(x, y) \\ & + w(1, 0)f(x + 1, y) + w(-1, 1)f(x - 1, y + 1) \\ & + w(0, 1)f(x, y + 1) + w(1, 1)f(x + 1, y + 1) \end{aligned}$$

The size $a \times b$ of a filter can be adjusted

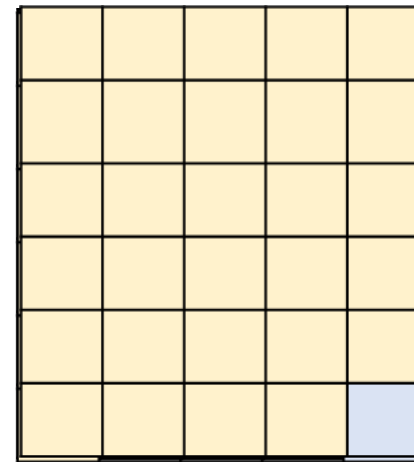
Spatial Correlation and Convolution

Correlation is the process of moving kernel over the image and computing the sum of products at each coordinate (x, y) .

- Moves the center of kernel over an image
- Compute the sum of products at each location
- Kernel of convolution is the rotated 180 degree of the correlation kernel.

0	0	0	0	0	0	0
0						0
0						0
0						0
0						0
0						0
0						0
0	0	0	0	0	0	0

Padding



Result of each center pixel $g(x, y)$

1	2	3
4	5	6
7	8	9

Kernel of Correlation

9	8	7
6	5	4
3	2	1

Rotated Kernel

Filter at each pixel $f(x, y)$

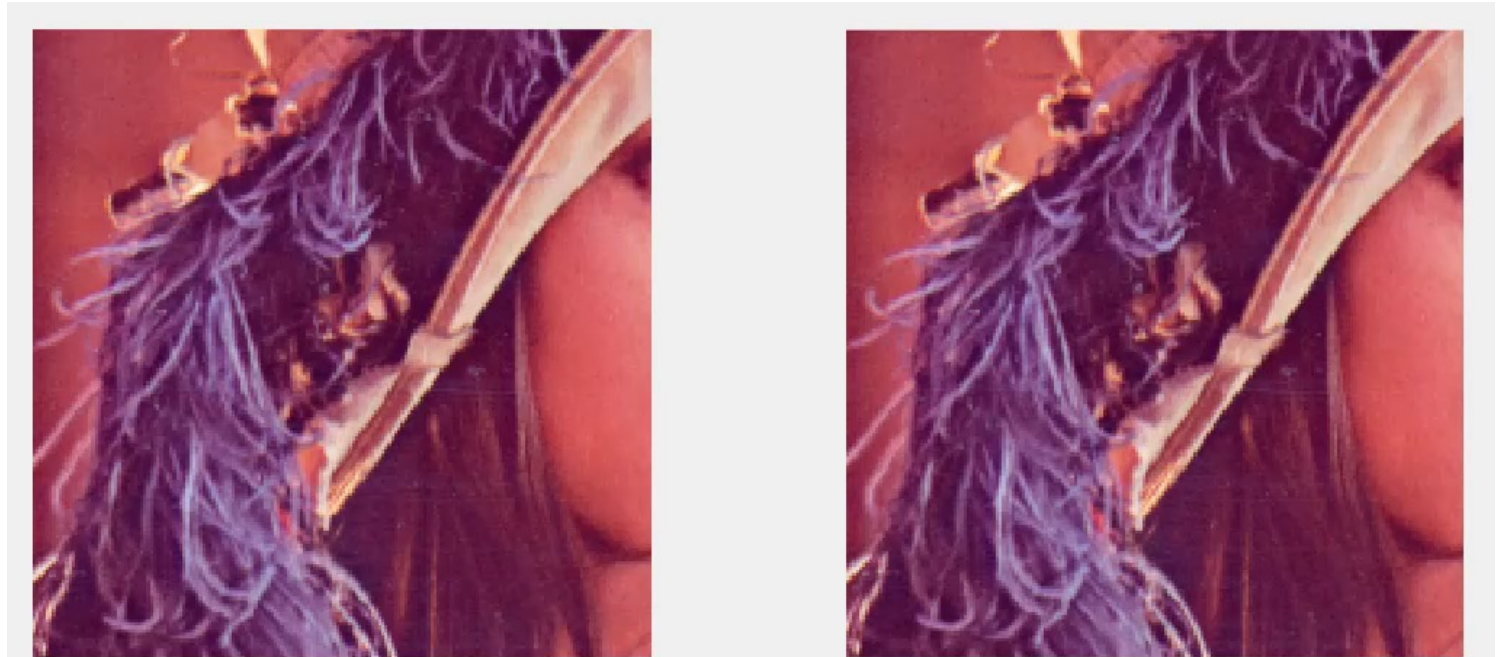
Spatial Filtering

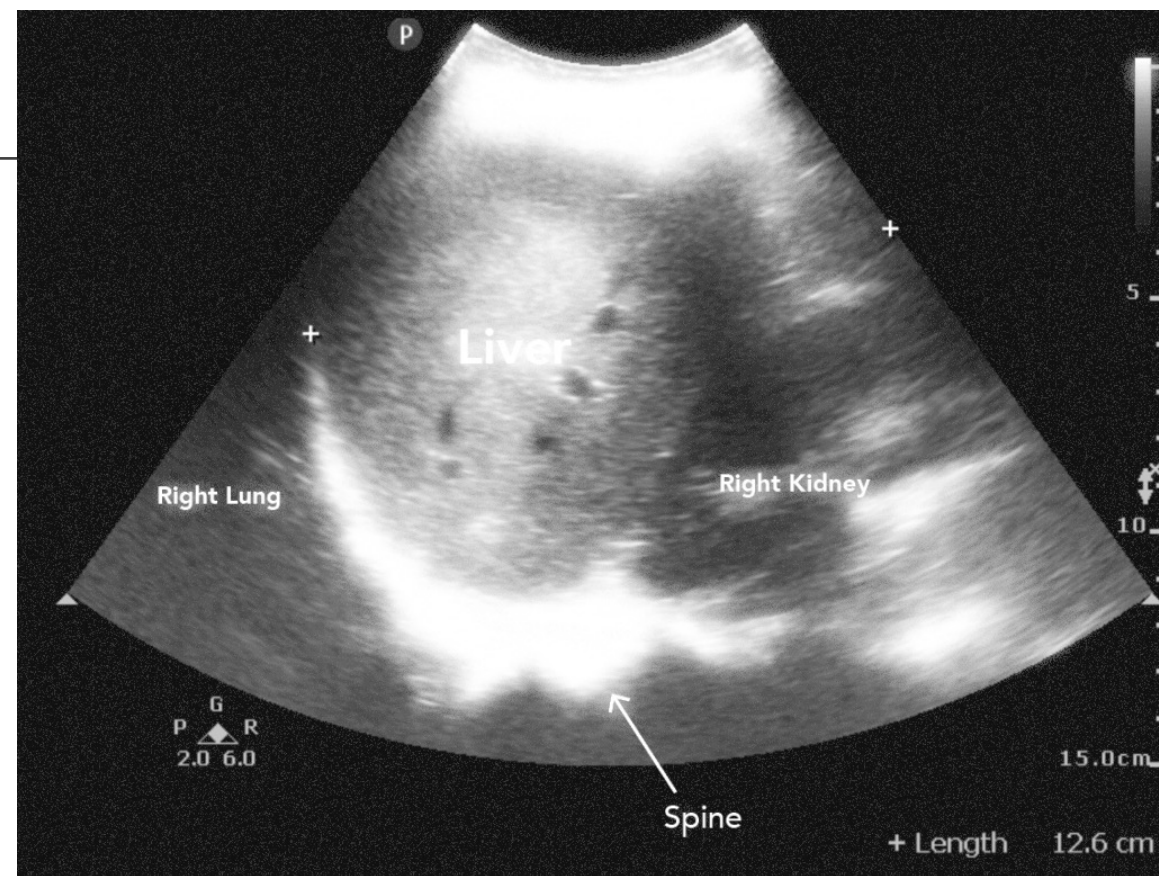
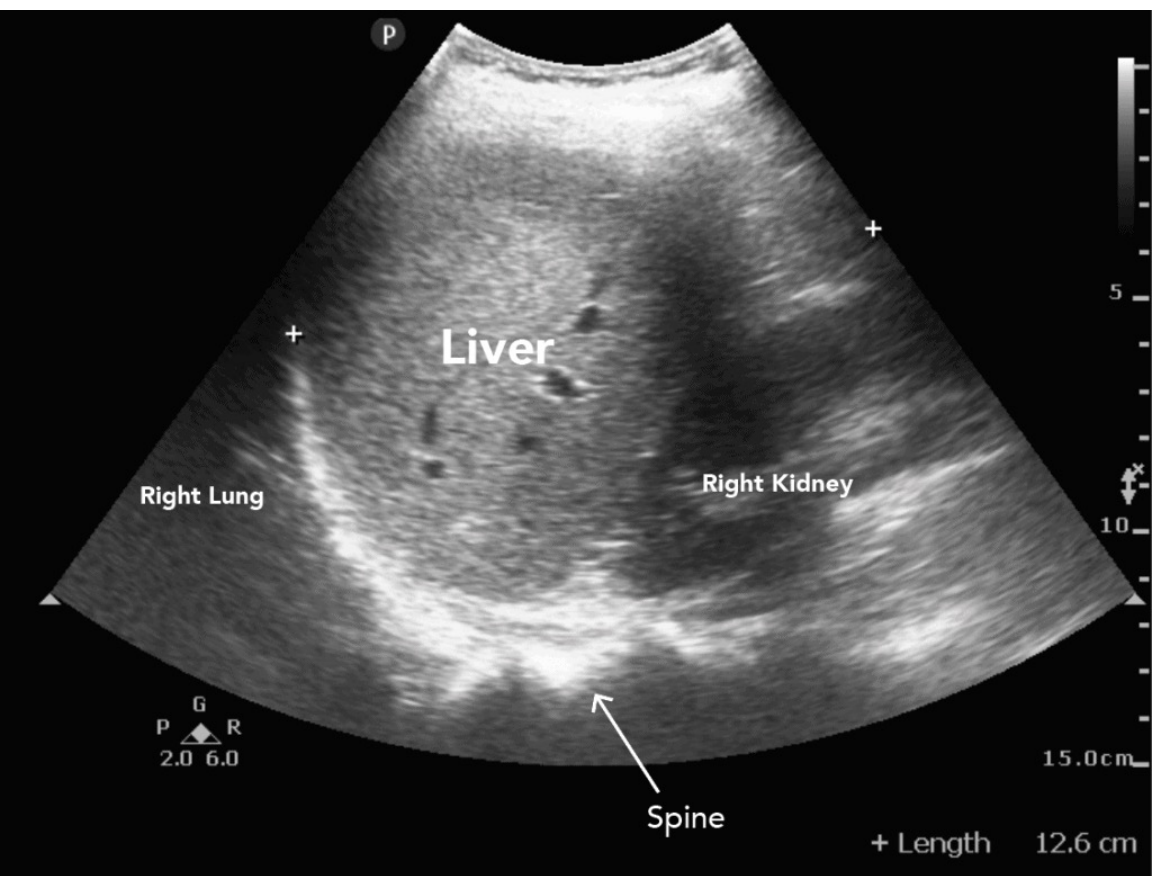
SMOOTHING FILTER

Smoothing Spatial Filters (Lowpass Filters)

Smoothing is applied for a noise reduction.

- Removing small irrelevant regions in an image.
- Smoothing contours.





Averaging Spatial Filtering

An averaging operator

Averaging spatial filters are used to reduce sharp transitions in intensity.

Given a filter of 1's of size $a \times b$, with a normalizing constant $\frac{1}{ab}$

$$\frac{1}{3 \times 3}$$

1	1	1
1	1	1
1	1	1

Filter of size 3×3

238	240	235	235	235	235	234
239	240	232	231	235	235	234
230	236	244	235	235	235	232
244	243	235	234	234	233	230
196	223	231	239	230	229	226
118	189	229	228	225	225	223
68	159	228	228	221	222	221
48	115	190	214	219	220	220

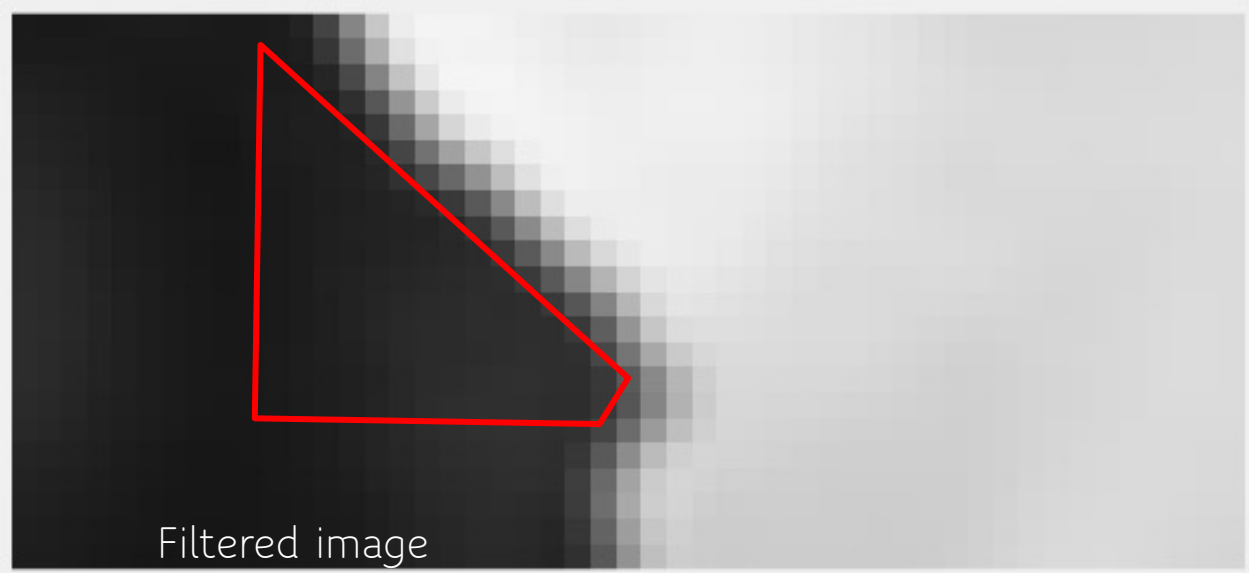
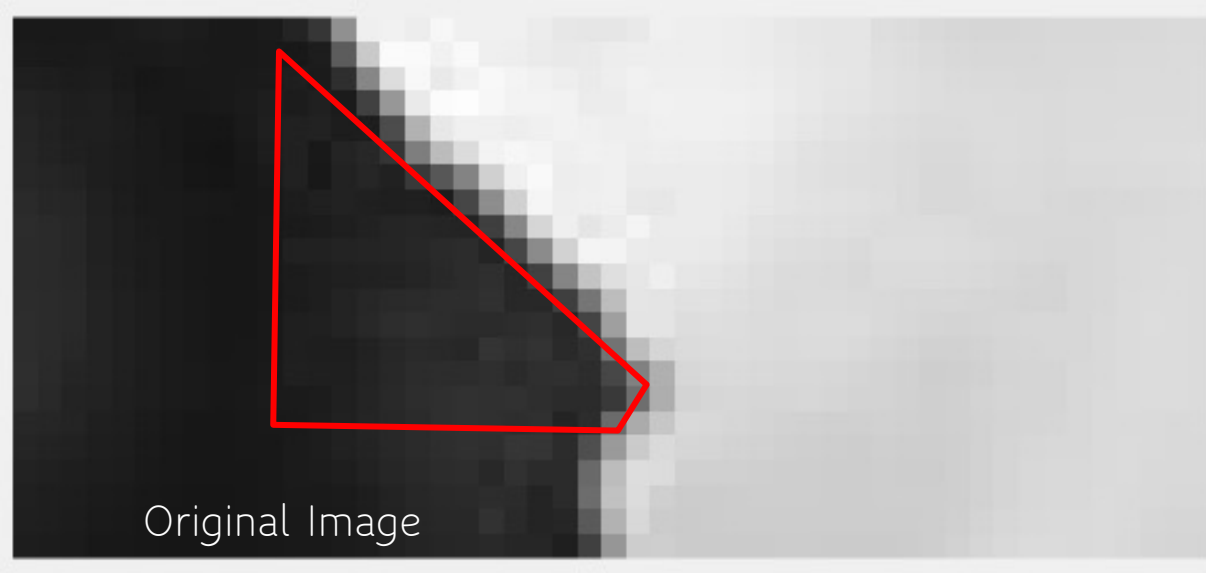
Original Image

241	239	238	236	235	235	235	234
237	237	237	236	235	235	234	233
222	235	238	237	235	234	234	231
186	217	231	236	235	234	232	229
134	180	212	228	232	231	228	226
86	132	182	217	229	227	225	223
57	94	149	198	220	222	222	221
48	71	117	168	202	215	219	219

Filtered image

Average intensity of constant intensity equals to the filtered image.

Averaging Spatial Filtering

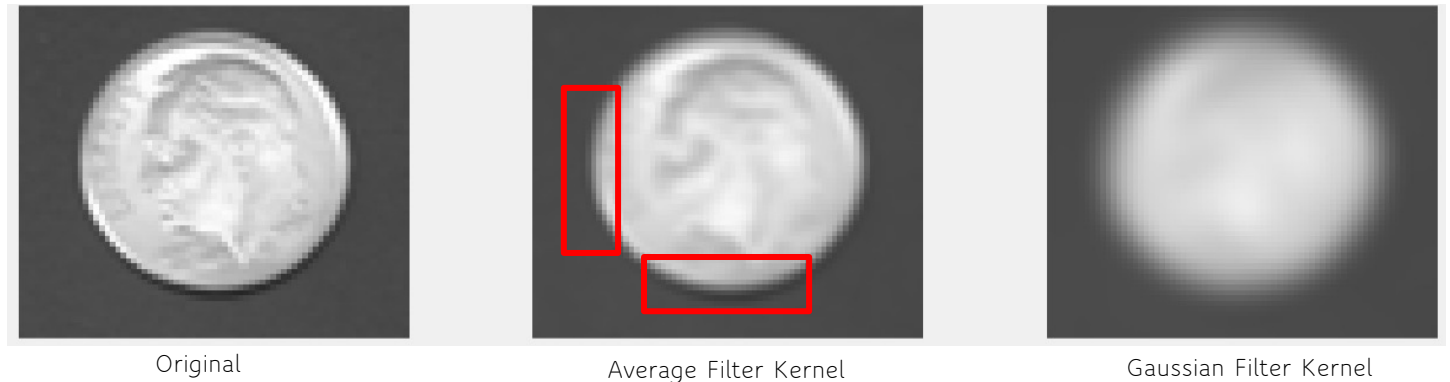


Gaussian Kernel

An averaging spatial filters applies the same weight to all pixels $f(x,y)$ in the kernel.

- It filters along perpendicular direction and gives an undesirable result for a detailed image or geometrical component.

Gaussian filter applies the values of the neighboring pixels are given different weighting



Gaussian Kernel

The weight of each neighboring pixels depends on the distance from the centering pixel $f(x,y)$.

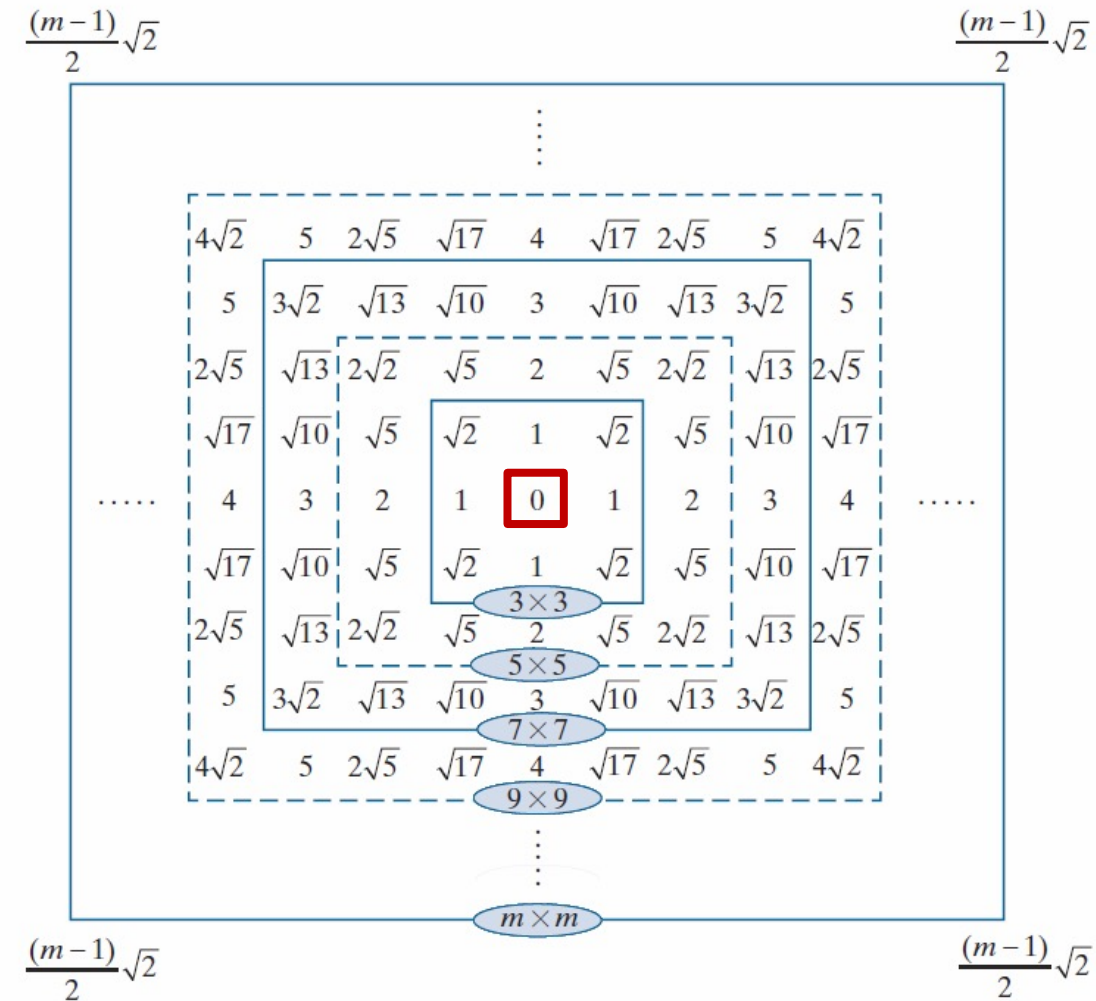
The form of Gaussian kernel is

$$w(s, t) = G(s, t) = K e^{\frac{-s^2 + t^2}{2\sigma^2}}$$

Let r be the distance from the center to any pixel $f(x + s, y + t)$ the variable becomes

$r = [s^2 + t^2]^{1/2}$ therefore

$$G(r) = K e^{\frac{r^2}{2\sigma^2}}$$



Smoothing Spatial Filters (Lowpass Filters)

The size of kernel to be applied for smoothing filters depend on the image's size.

- Applying a large size of kernel to a small image produces a blurred image.

Smoothing is a step that often used before extracting an interested region on the image.

- To obtain a smooth boundary

- To avoid extracting noised region

- To facilitate a measurement of region with a nonuniform illumination.



a b c

FIGURE 3.41 (a) A 2566×2758 Hubble Telescope image of the *Hickson Compact Group*. (b) Result of lowpass filtering with a Gaussian kernel. (c) Result of thresholding the filtered image (intensities were scaled to the range $[0, 1]$). The Hickson Compact Group contains dwarf galaxies that have come together, setting off thousands of new star clusters. (Original image courtesy of NASA.)

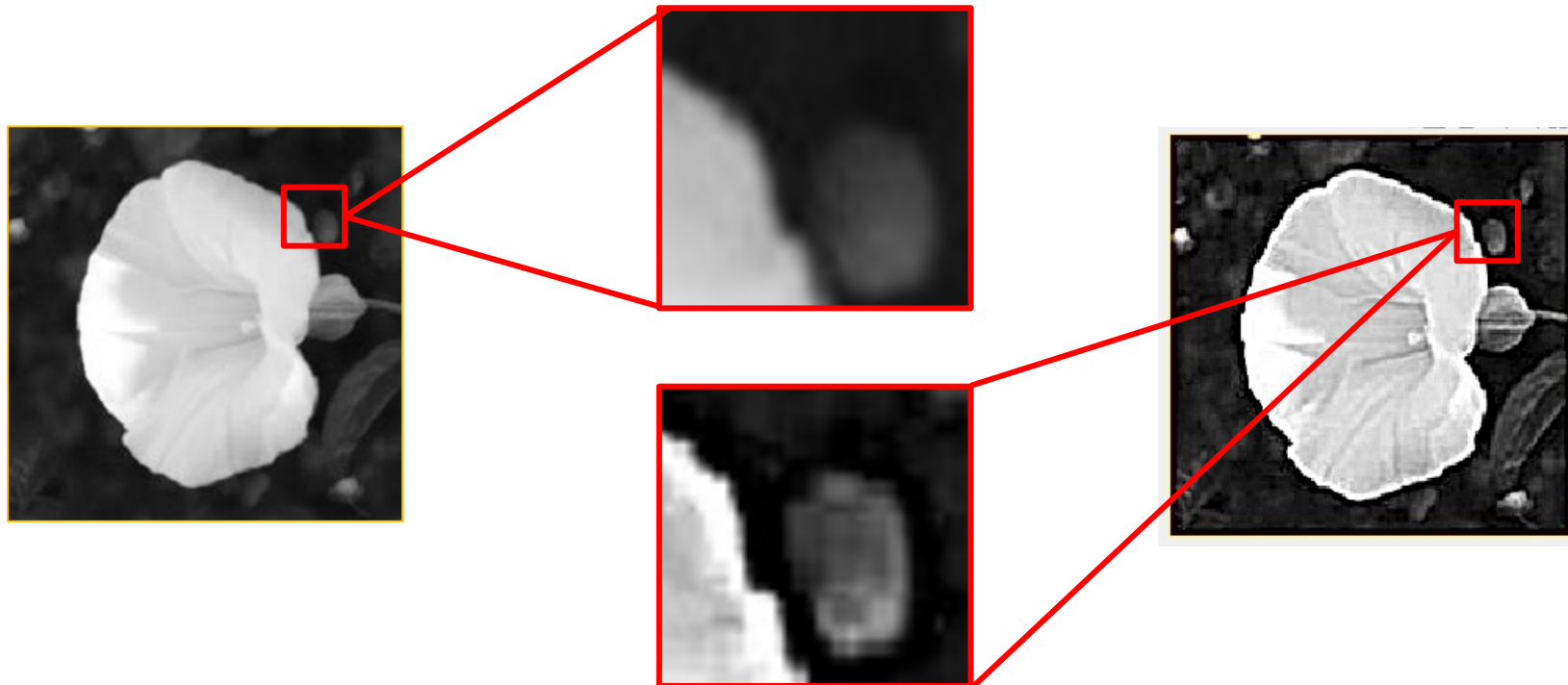
SHARPENING FILTER

Spatial Filtering

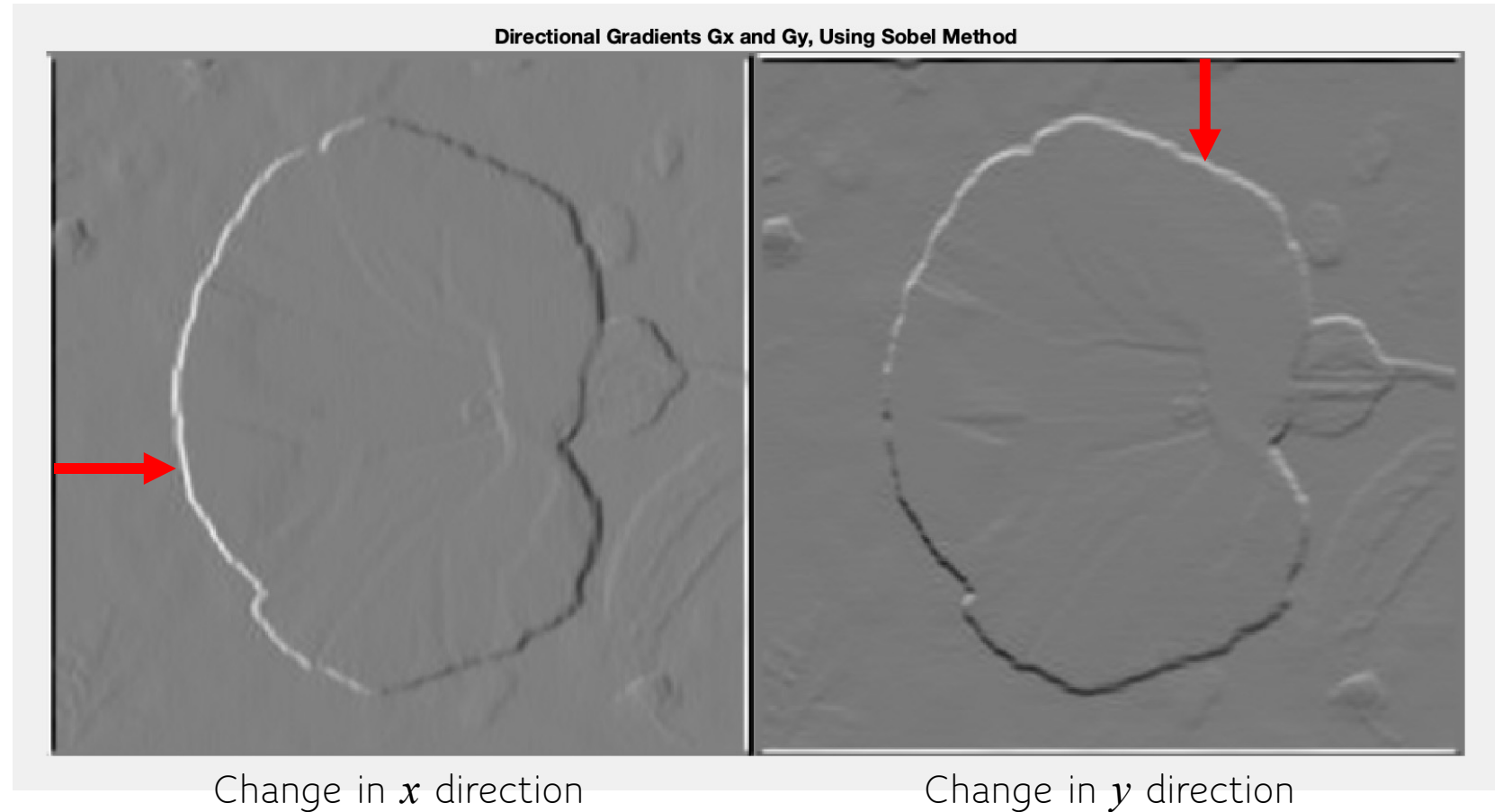
Sharpening Spatial Filters (Highpass Filters)

Sharpening is applied for highlighting transition in intensity.

- Enhance edges and discontinuities regions in an image.



- The discontinuities of region is implemented using the magnitude of the **gradient**.
 - An **image gradient** is a directional change in the intensity or color in an image
-



Gradient Operator

Similar to an averaging operator for smoothing or noise removal.

It uses neighboring pixels for the operation.

The weights of the filter contain both positive or negative value.

<i>-1</i>	<i>1</i>
-----------	----------

Change in *x*-direction

<i>1</i>
<i>-1</i>

Change in *y*-direction

<i>10</i>	<i>5</i>	<i>5</i>
<i>10</i>	<i>40</i>	<i>40</i>
<i>10</i>	<i>40</i>	<i>40</i>

A 3 × 3 region of an image

<i>0</i>	<i>-35</i>	<i>-35</i>
<i>0</i>	<i>0</i>	<i>0</i>
<i>10</i>	<i>40</i>	<i>40</i>
<i>0</i>	<i>0</i>	<i>0</i>

Result of *y*-direction

- The discontinuities of region is implemented using the magnitude of the **gradient**.
- An **image gradient** is a directional change in the intensity or color in an image

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Horizontal or vertical direction

Diagonal direction

- The magnitude of vector at (x, y) is the rate of change in the direction of gradient vector.
- Denoted as $M(x, y)$, where $M(x, y) = \sqrt{g_x^2 + g_y^2}$

Gradient Operator

-1	1
------	-----

Change in x -direction

5	5	5
5	40	40
5	40	40

A 3×3 region of an image

0	5	5	5	0
0	5	40	40	0
0	5	40	40	0

Padded image

5	10	10	5
5	45	80	40
5	45	80	40

Gradient in x -direction (g_x)

1
-1

Change in y -direction

0	0	0
5	5	5
5	40	40
5	40	40
0	0	0

Padded image

5	5	5
10	45	45
10	80	80
5	40	40

Gradient in y -direction (g_y)

14.2	46.1	45.3
46.1	113.2	89.5
45.3	89.5	56.6

An image gradient

$$M(x, y) = \sqrt{g_x^2 + g_y^2}$$

Other Operators

Sobel Operator

- It returns large response on vertical or horizontal changes.

x -direction (g_x)			y -direction (g_y)		
1	0	-1	1	2	1
2	0	-2	0	0	0
1	0	-1	-1	-2	-1

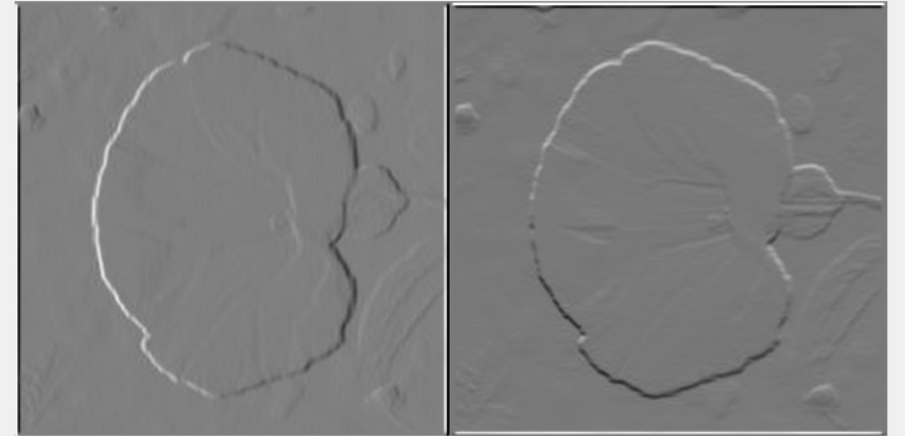
Sobel Operator

Prewitt Operator

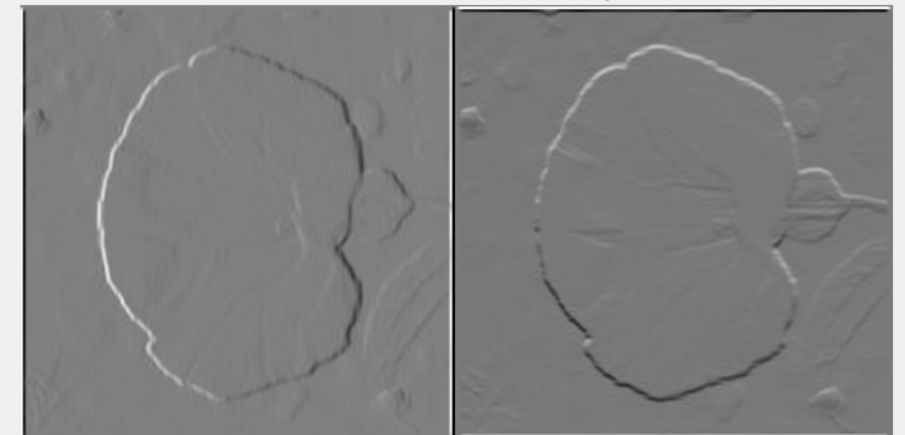
x -direction (g_x)			y -direction (g_y)		
1	0	-1	1	1	1
1	0	-1	0	0	0
1	0	-1	-1	-1	-1

Prewitt Operator

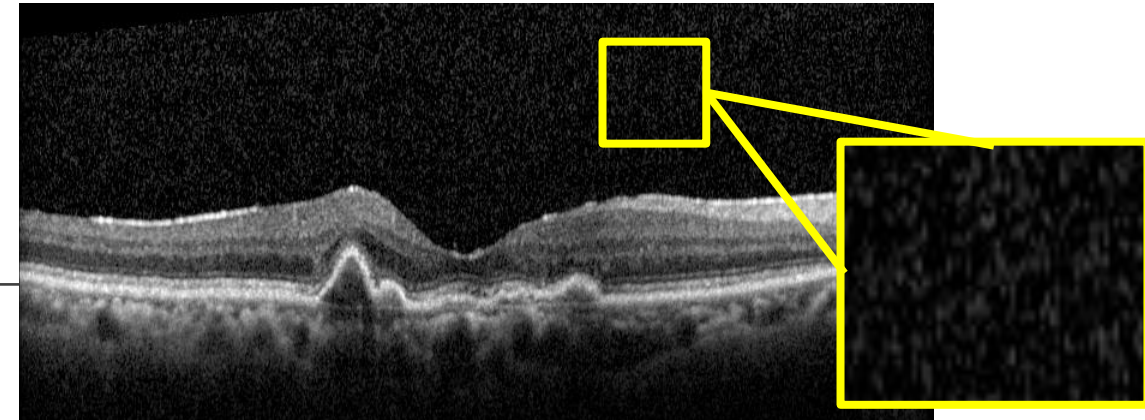
Directional Gradients Gx and Gy, Using Sobel Method



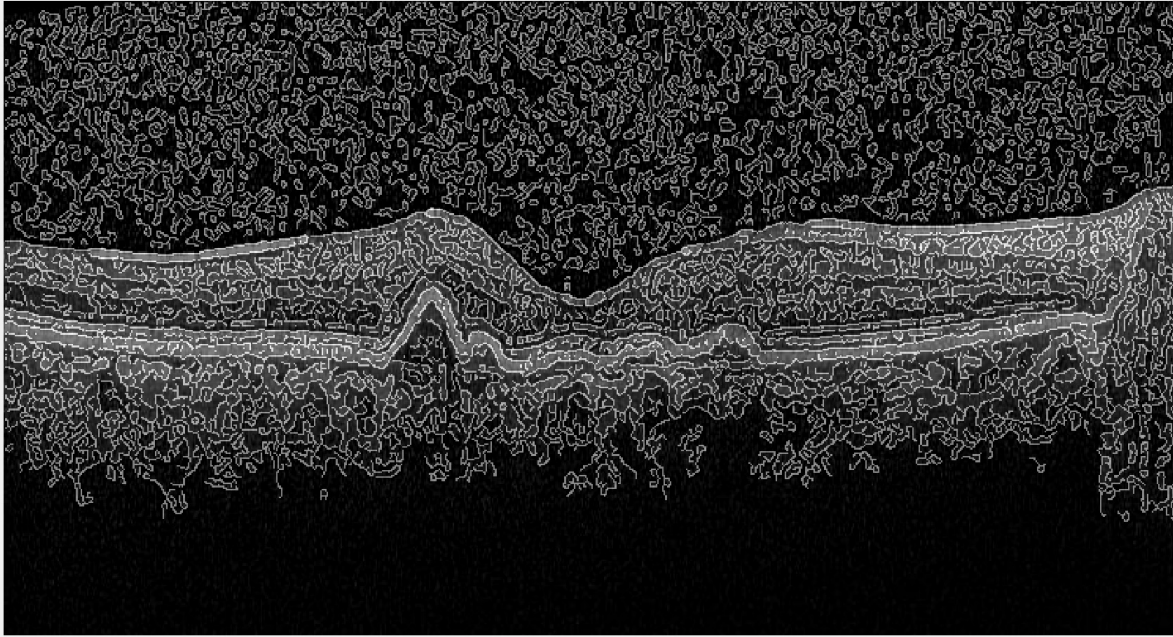
Directional Gradients Gx and Gy, Using Prewitt Method



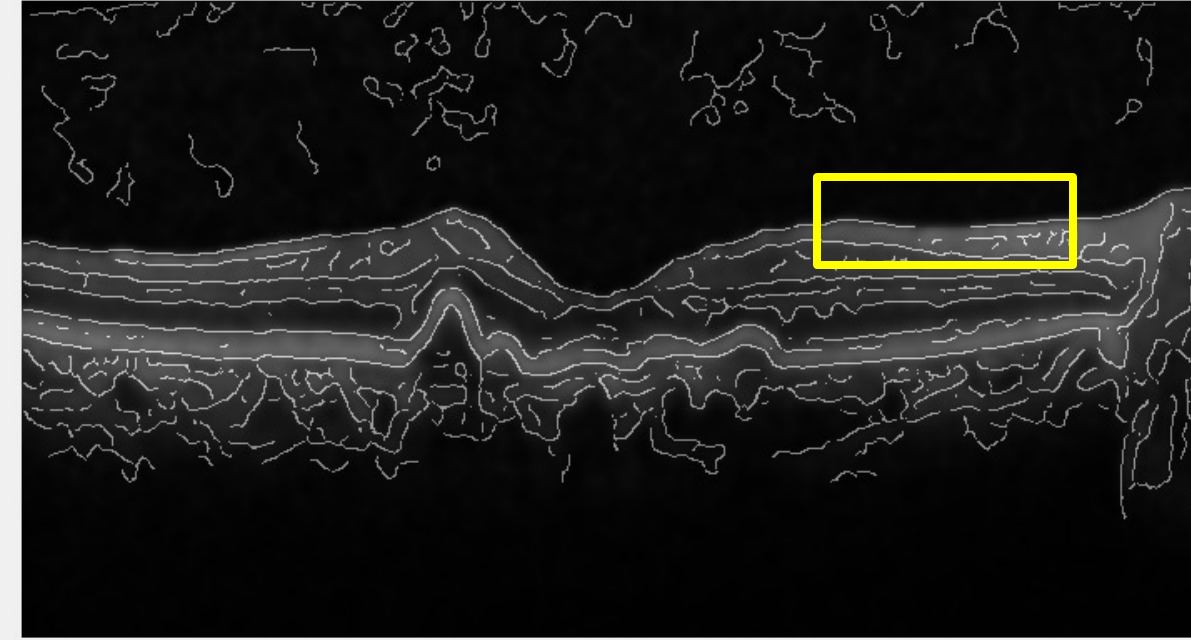
Sharpen and Smoothing Filters



Contour of Sharpen Image

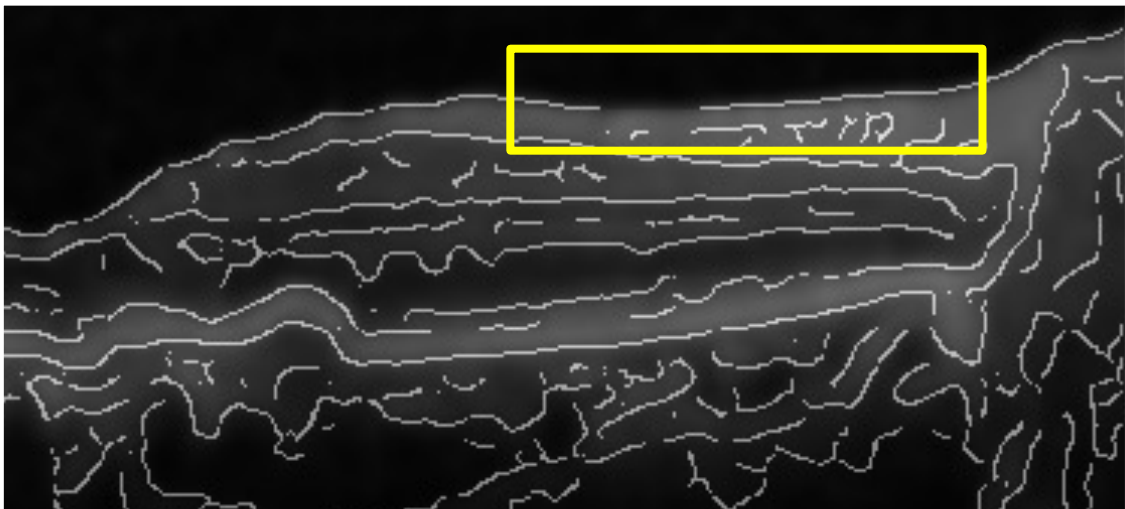
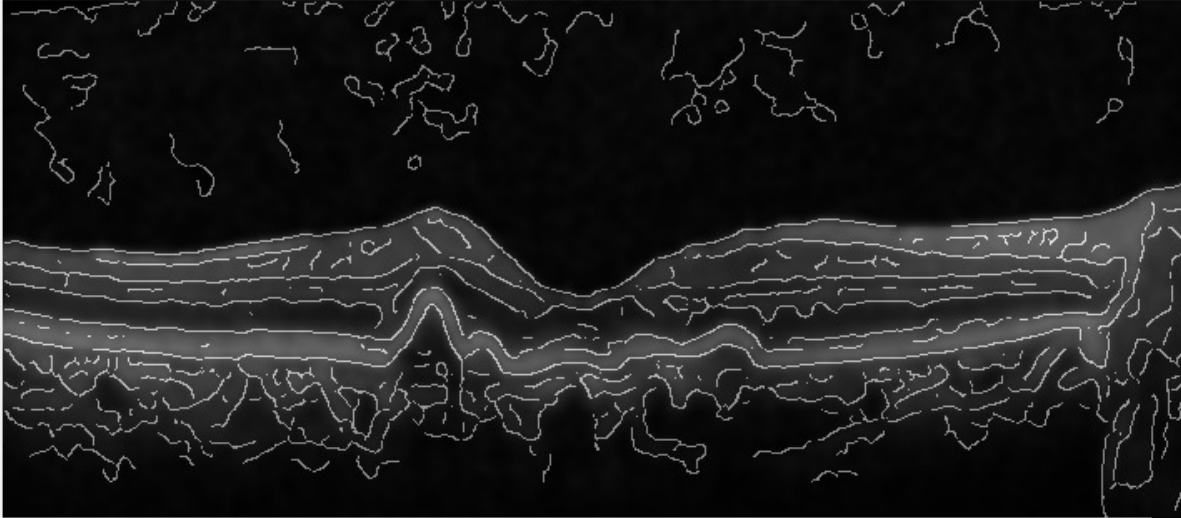


Contour of Smoother Image

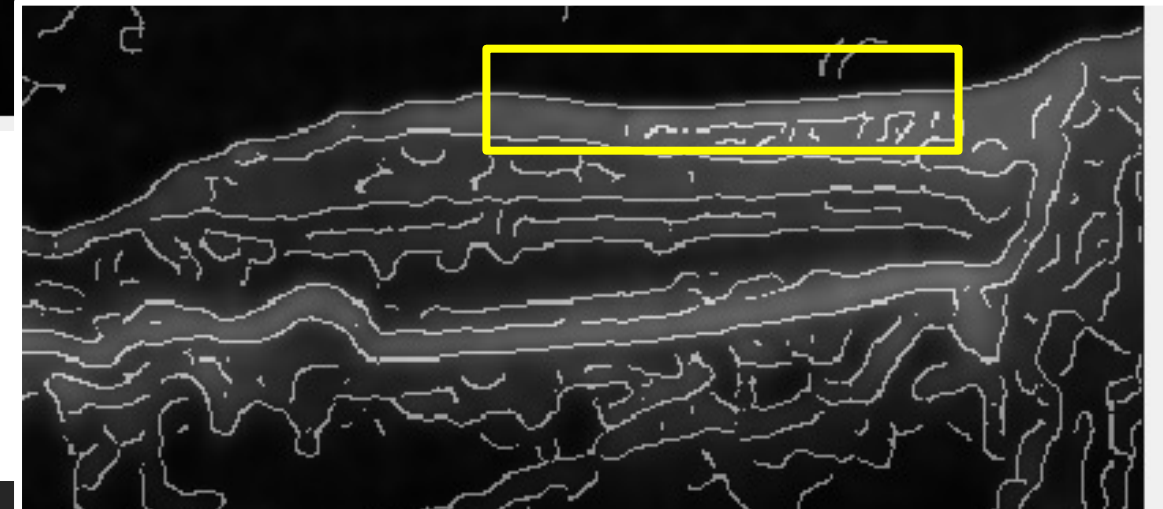
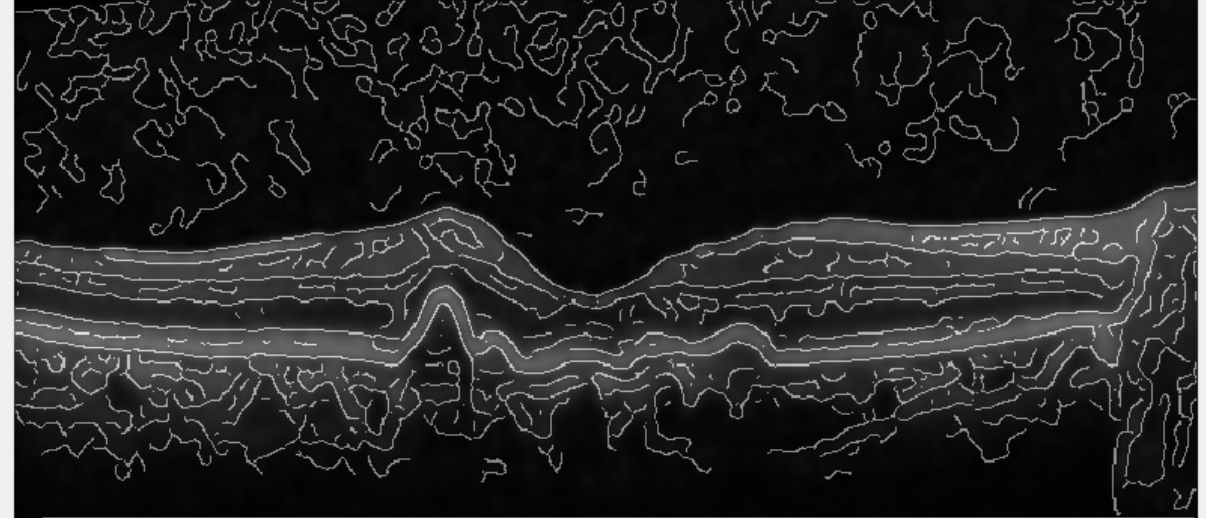


Combine Smoothing and Shapening Filters

Contour of Smoothen Image



Contour of Smoothen-Sharpen Image



Agenda



**01. Imaging
Fundamental**
Image Acquisition

02. Medical Imaging
Types

**03. Fundamental of
Image Processing**
Basic processing
Spatial Filter

**04. Morphological
Technique**
Morphology

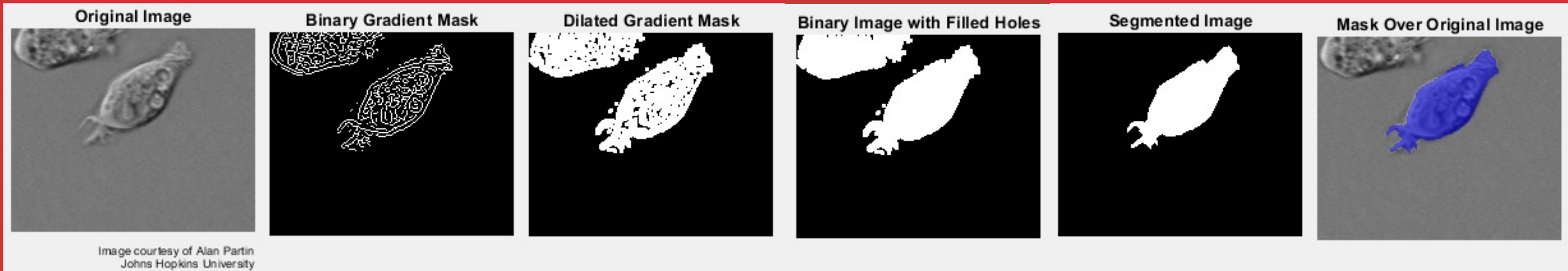


04.

Morphological Technique

Morphology

- Morphology is used for extracting image's components.
 - Simplify object's structure and facilitate a representation in mid-level image processing.



Binary Image

- To apply morphology,
 - The input image is a binary image
 - The pixel value $f(x, y) = 1$ is foreground
 - The pixel value $f(x, y) = 0$ is background

Original Image

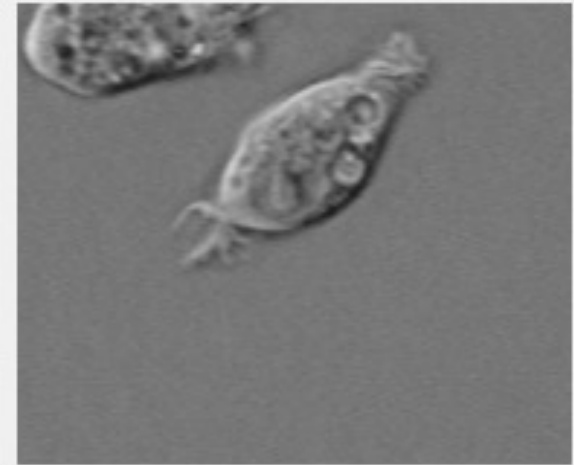
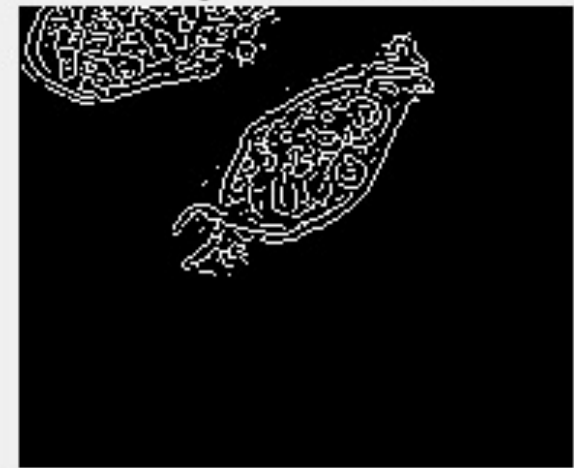


Image courtesy of Alan Partin
Johns Hopkins University

Binary Gradient Mask



Morphology

(1,1)

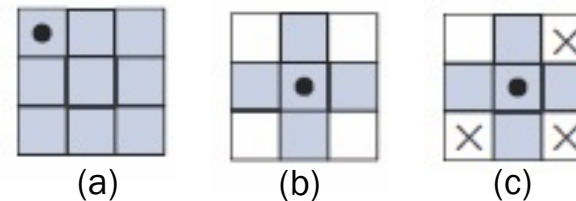
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	0
0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0
0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0
0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

(17,17)

A digital image

- A set theory of two types:
- A digital image (objects)
 - Each set is coordinates of an object in a digital image.
 $A = \{ (8, 2), (6, 2), (7, 2), (8, 2), \dots, (8, 15), (9, 15) \}$
- A digital structure element (structuring element)
 - Structure of an object (kernel)
 - Set of structure element (b)

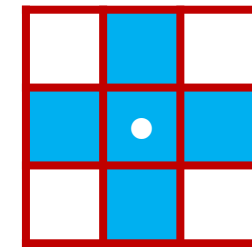
$$B = \{ (-1, 0), (0, -1), (0, 0), (0, 1), (1, 0) \}$$



A structure element

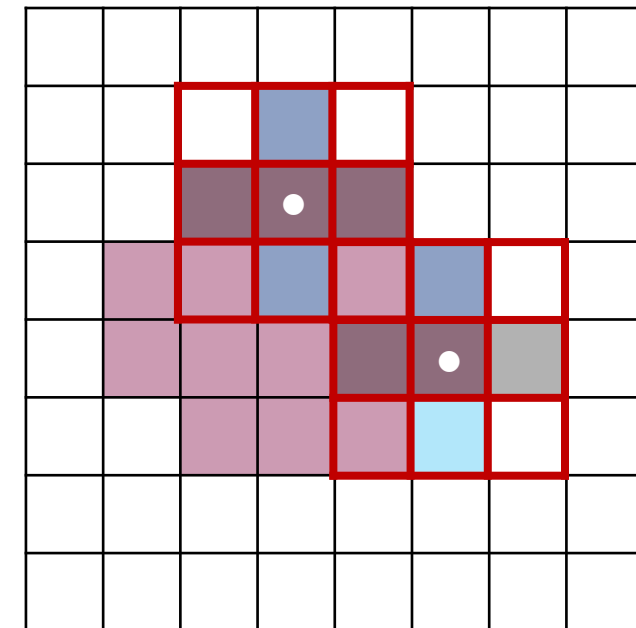
Morphology

- The object A and the structure element B are used to form a new image of the same size as the input image I



Structure element B

- To form a new image
 - The structure element B is applied over the image I
 - If the structure element B is completely contained A then
 - Mark the location as a foreground.
 - Otherwise, mark the location as a background

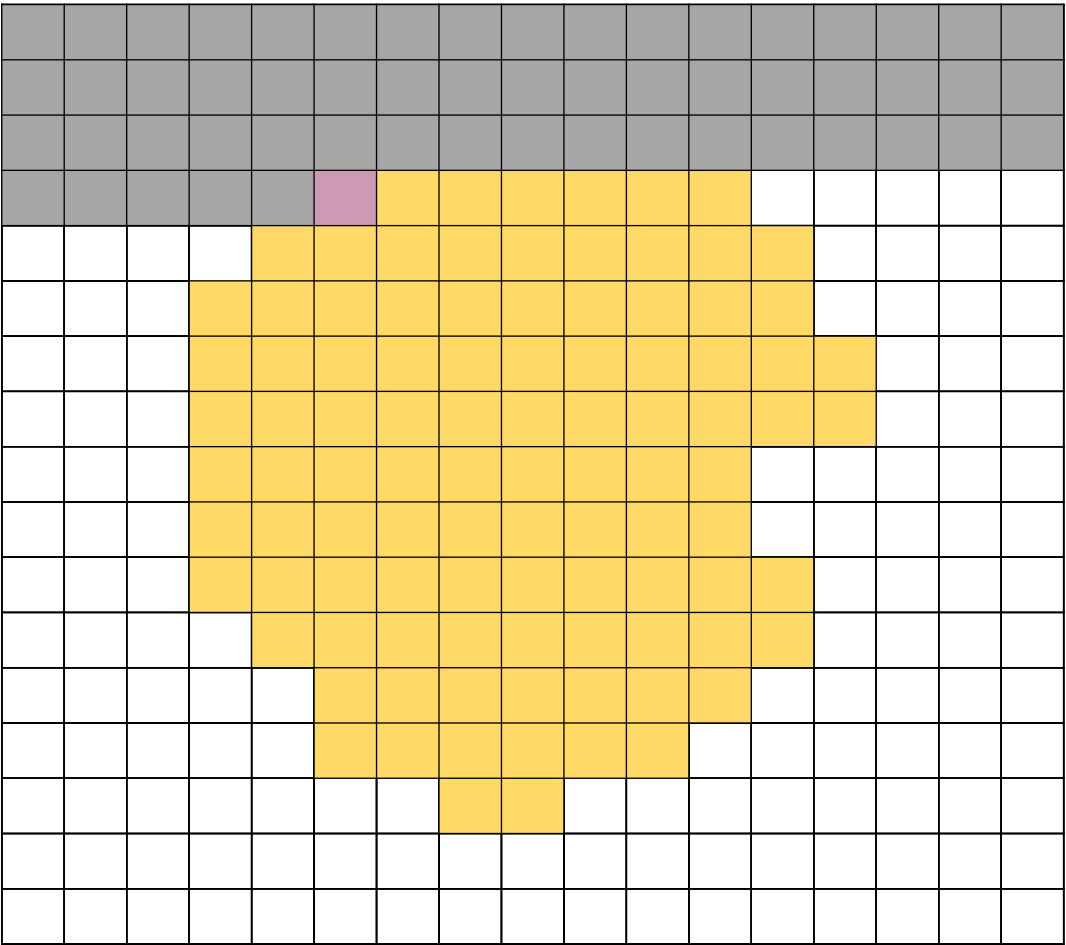


Object A

A structure element *B*

	●	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	●	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	0
	0	0	0	0	0	1	●	1	1	1	1	1	0	0	0	0	0
	0	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0
	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	0
	0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	0
	0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	0
	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0	0
	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0	0
	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0
	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0
	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

A digital image *I*



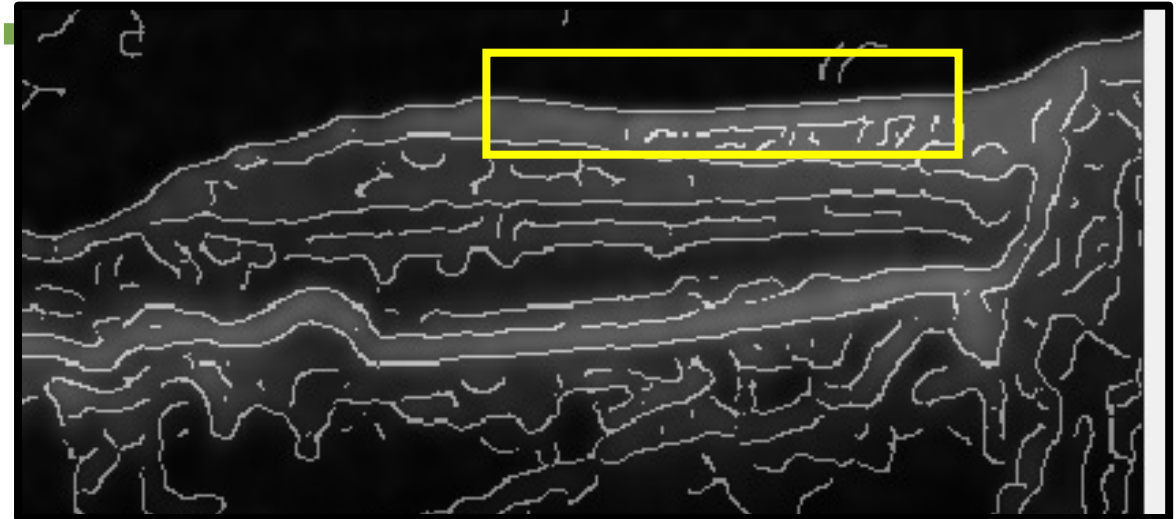
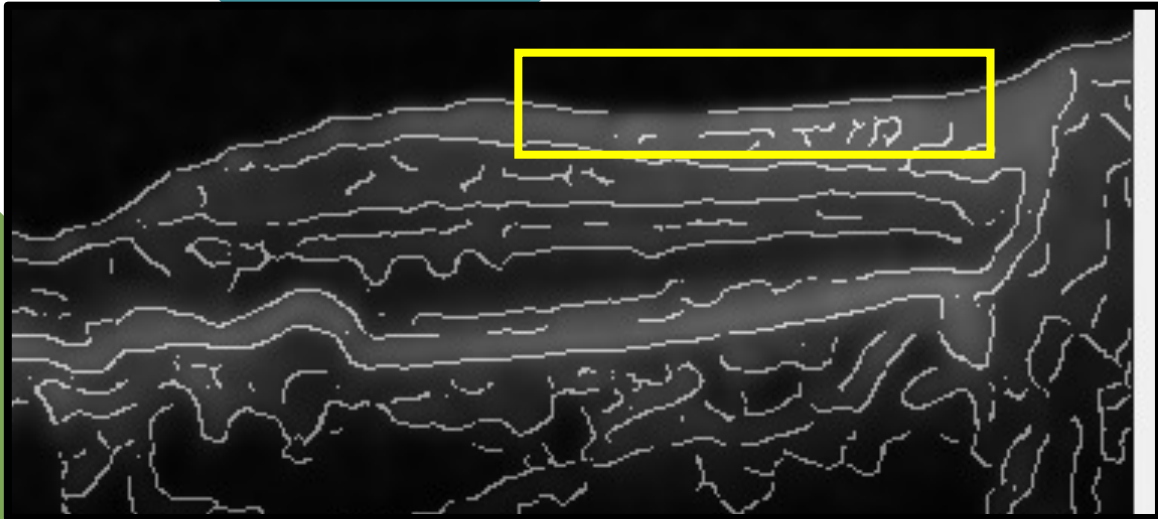
[illegible]

A digital image I

[illegible]

Transformed digital image I using the structure B

Erosion and Dilation



Erosion

- An **operator** in the area of mathematical morphology.

$$A \ominus B = \{z | (B)_z \subseteq A\}$$

- The set of all points z such that B is contained in A
- **Input:**
 - An image I which is to be eroded, denoted as A .
 - A set of coordinate points known as a structuring element B
 - It determines the precise effect of the erosion on the input image.

[illegible]

A set of object A

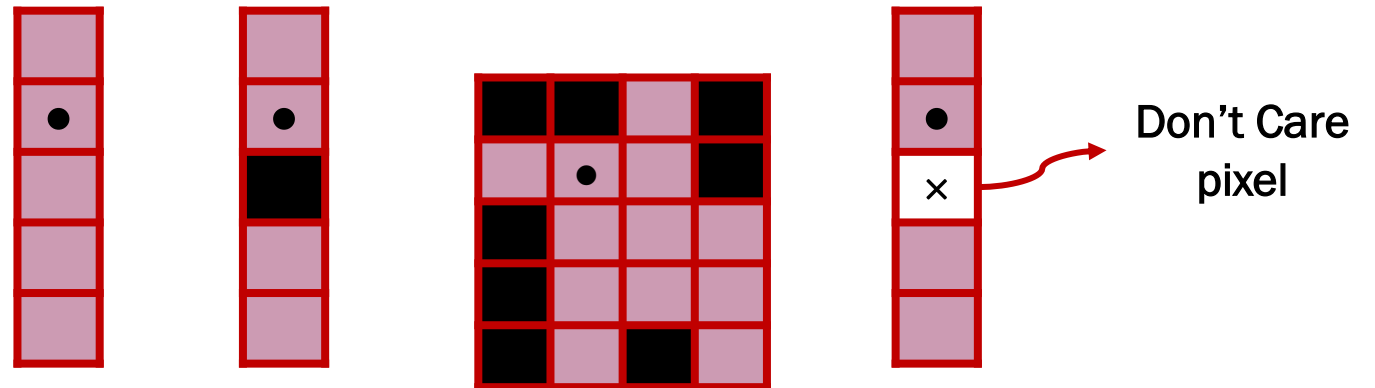
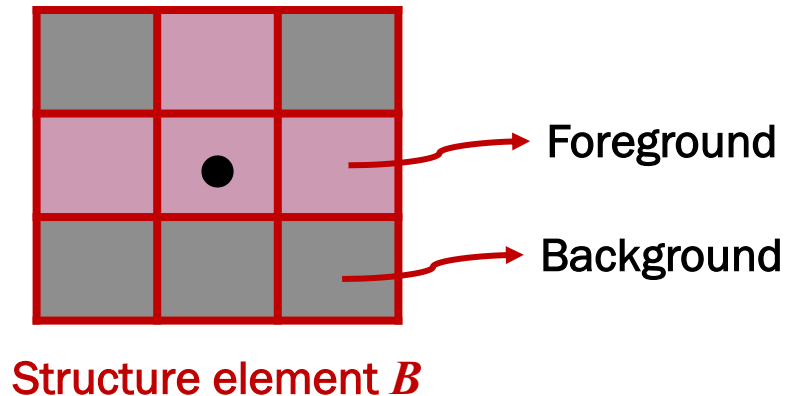
	●	

[illegible]

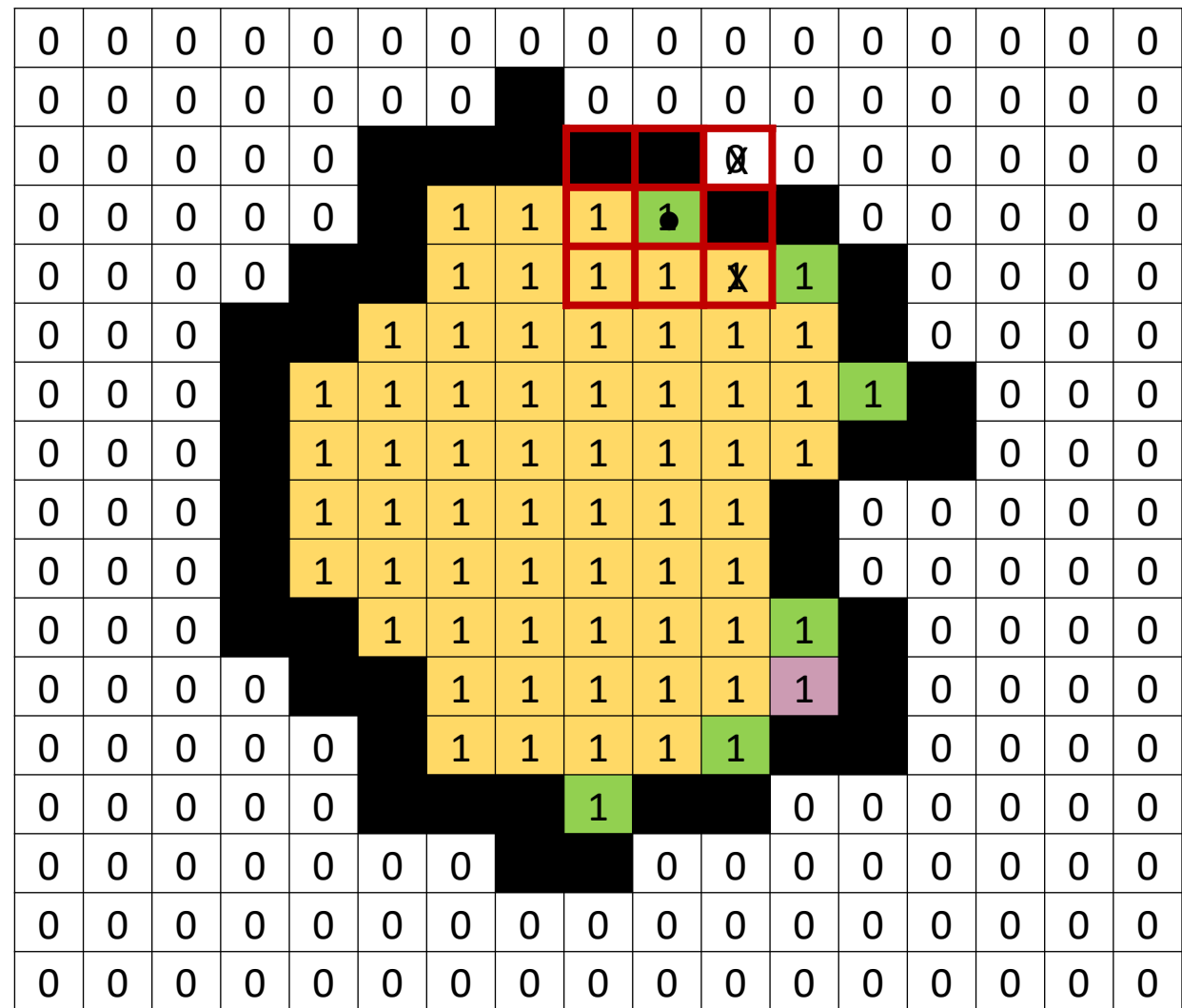
Erosion of object A
using the structure B ,

Erosion – Structure Element

- Erosion is to remove foreground pixels of the image.
- The structure element of a structure element B affects the output of erosion.
- The structure element can contain both foreground and background pixels



Example of Structure element B

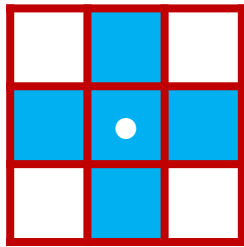


Dilation

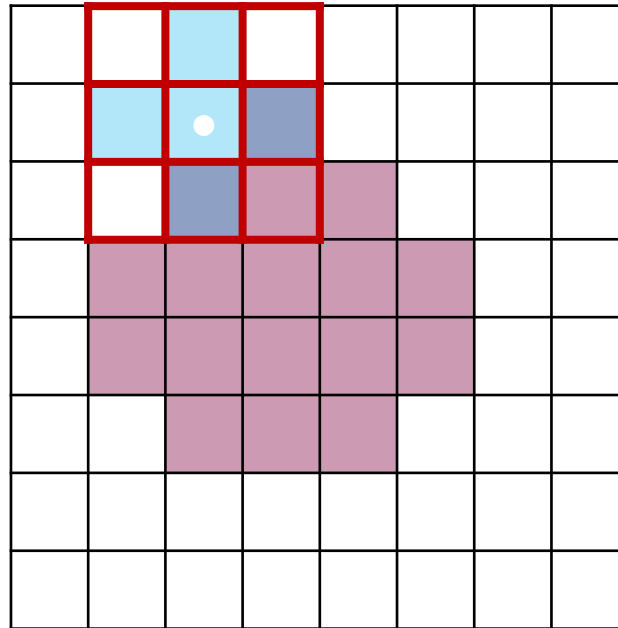
- An operator in the area of mathematical morphology.

$$A \oplus B = \{z | (B)_z \cap A \neq \emptyset\}$$

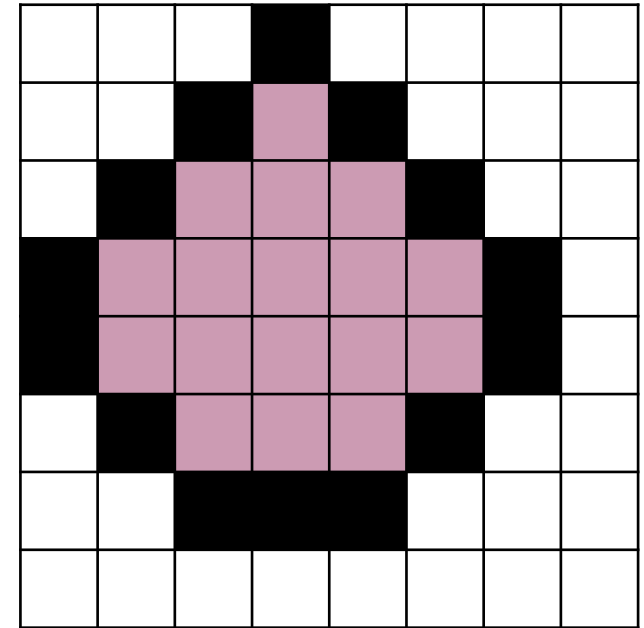
- The set of all points z such that B is overlapped at least one element of A



Structure element B



Object A

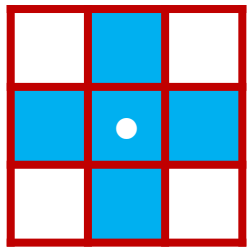


Erosion VS Dilation

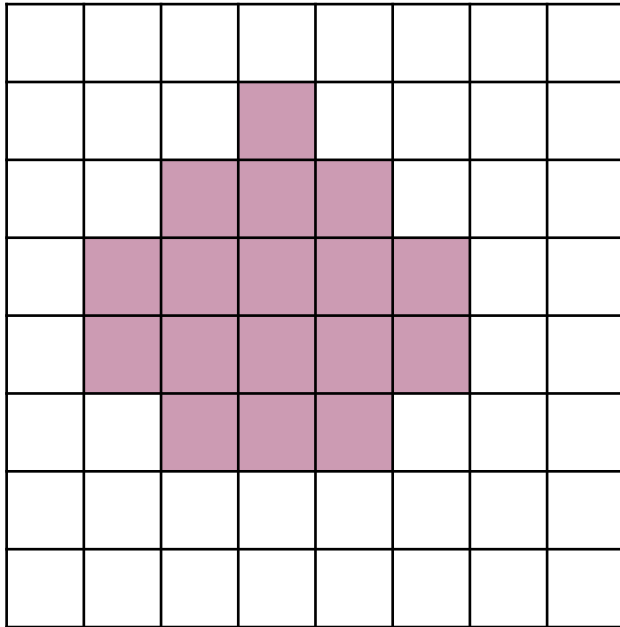
- Erosion:

- Shrinking or thinning operation
- Every pixels in the structure element must be in the foreground.

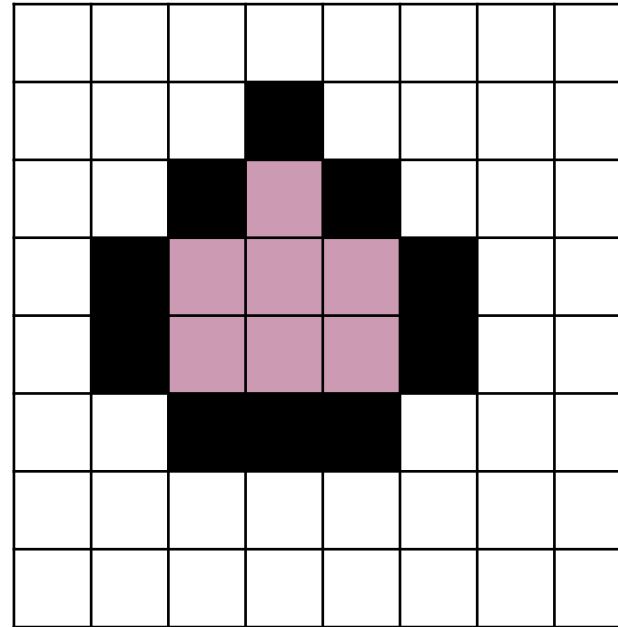
- Growing or thickening operation
- At least one pixel in the structure element is overlapped with the foreground.



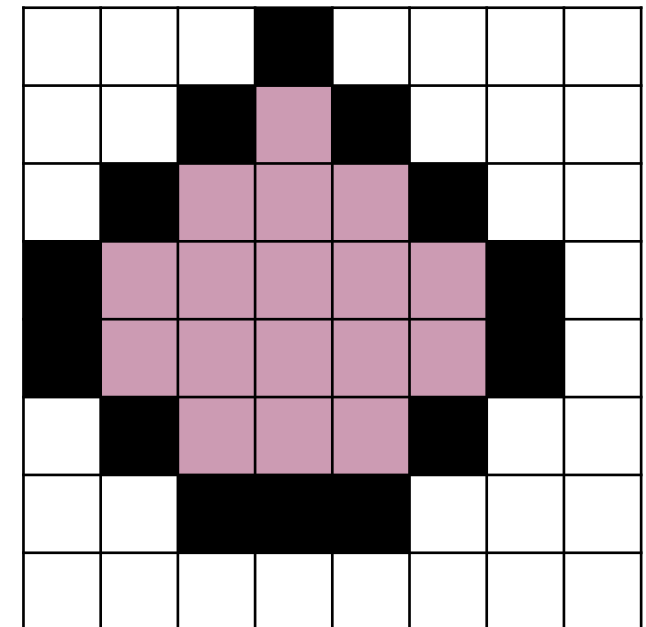
Structure
element *B*



Object *A*

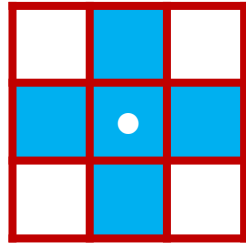


Result of erosion



Result of dilation

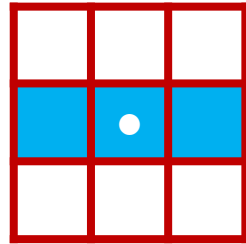
Erosion



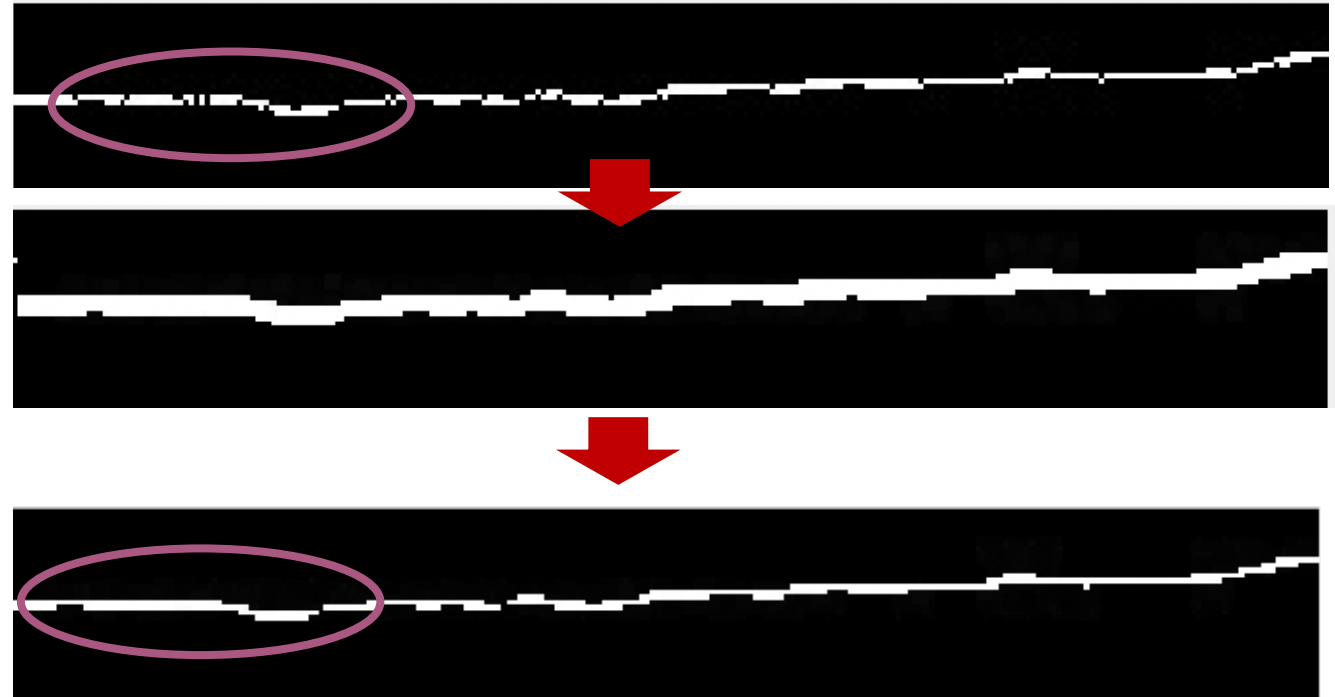
Structure element B



Dilation



Structure element B



Dilation → Erosion

Agenda

**01. Imaging
Fundamental**
Image Acquisition

02. Medical Imaging
Types

**03. Fundamental of
Image Processing**
Basic processing
Spatial Filter

**04. Morphological
Technique**
Morphology

05. Ultrasound
Abdomen



Recap Image Processing

Intensity Transformation

Slicing

Histogram Equalization

Spatial Filtering

Sharpen

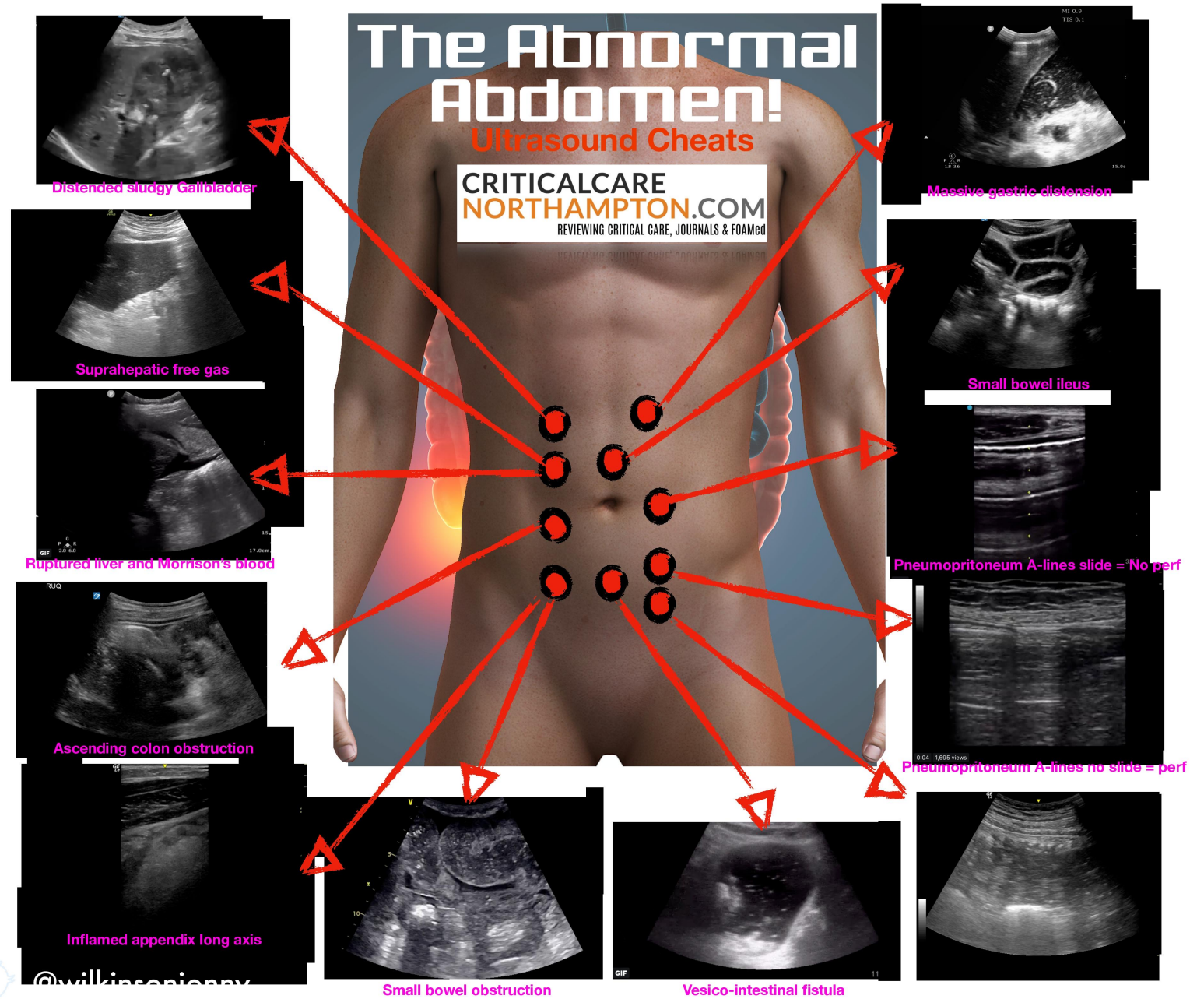
Smoothen

Morphological

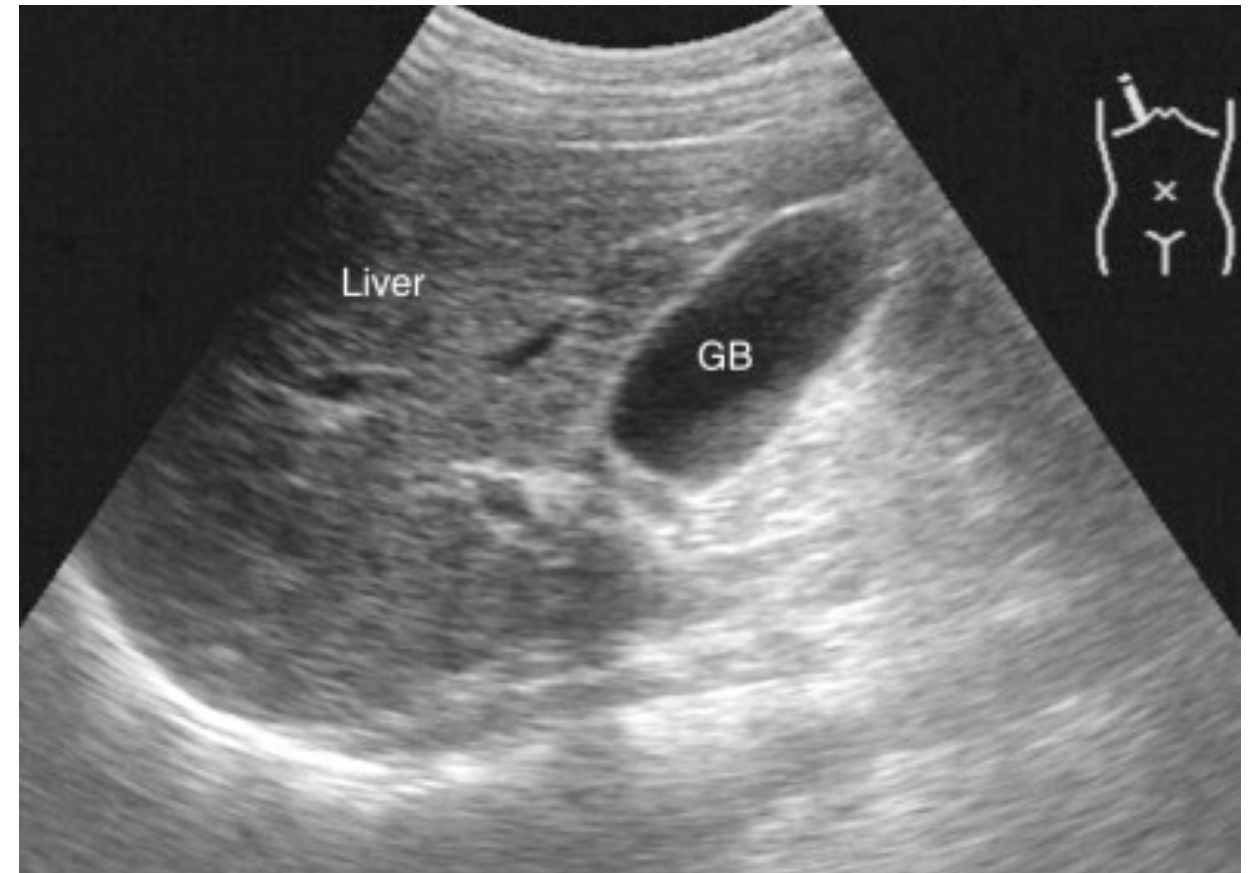
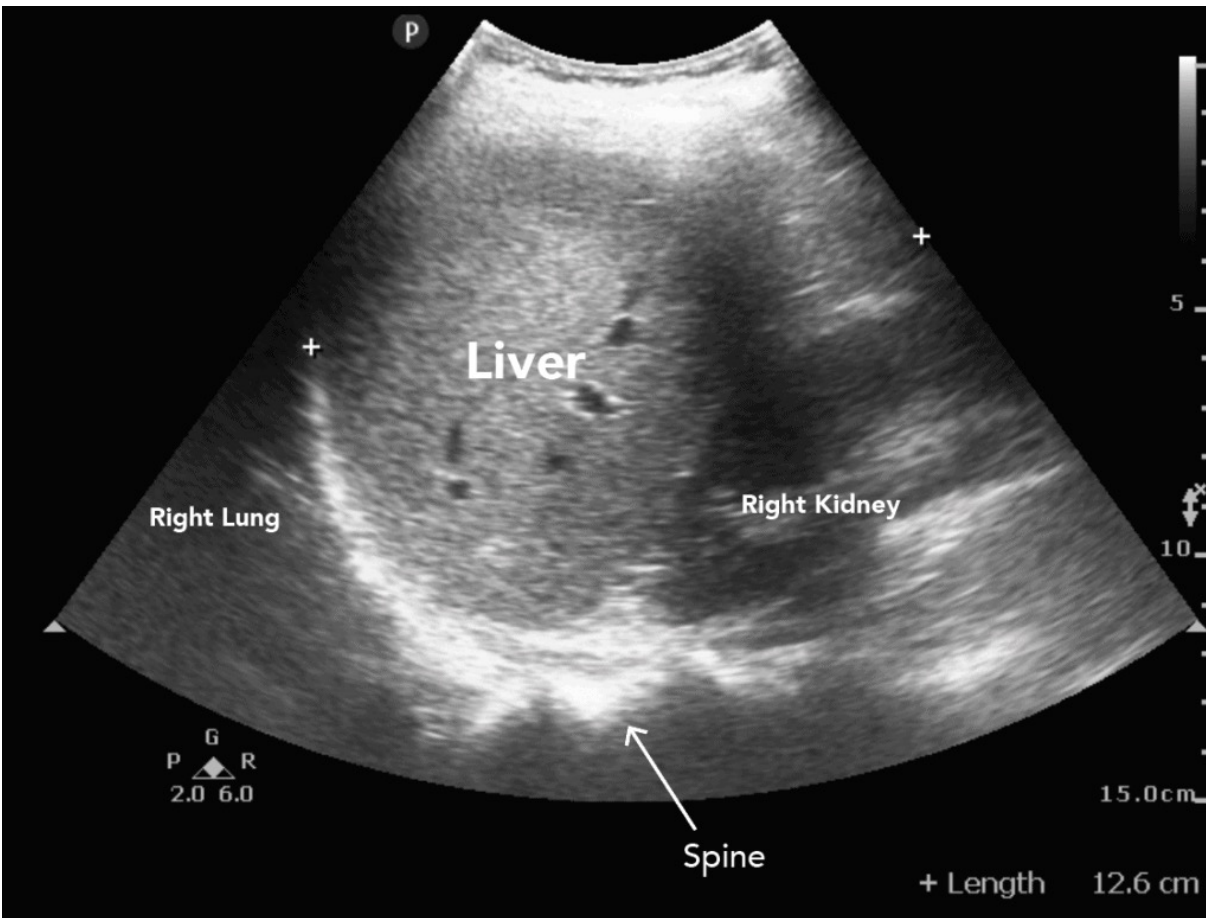
Erosion

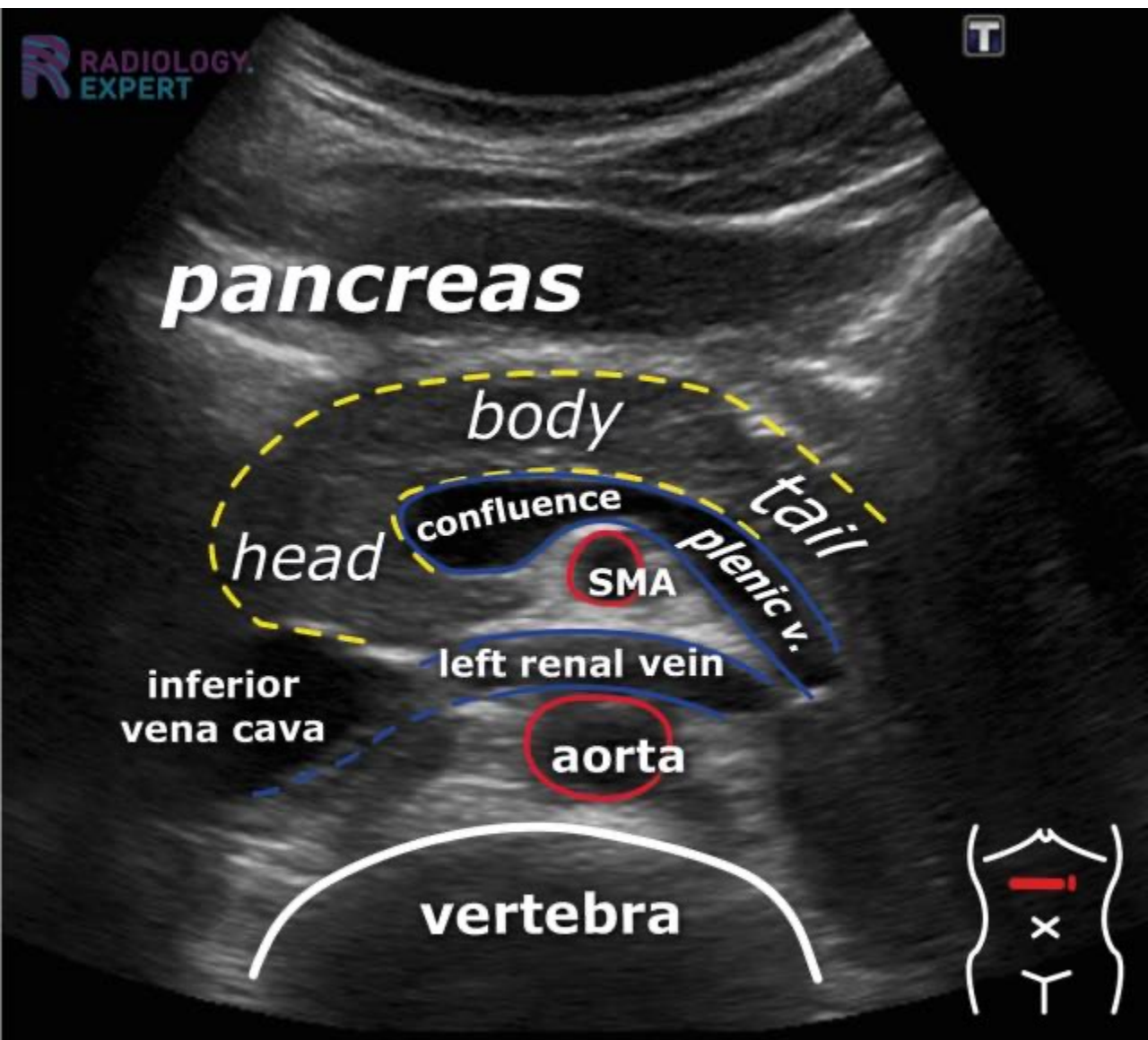
Dilation

Difficulty of Abdomen Ultrasound

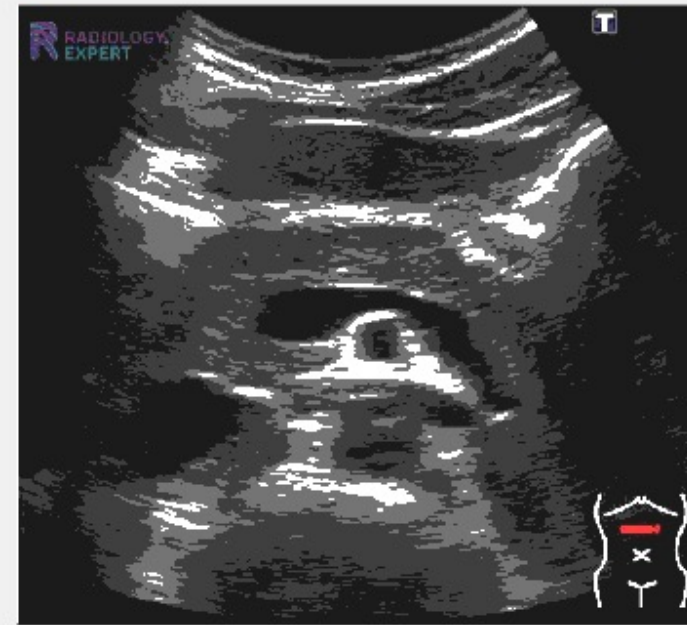
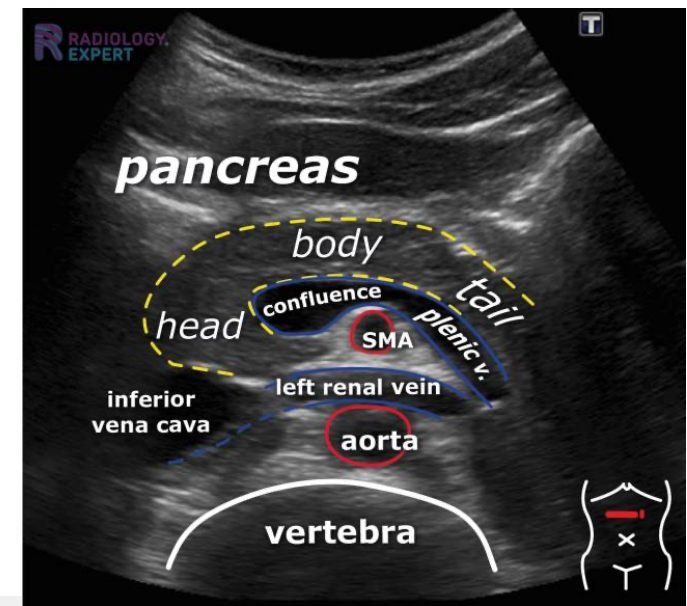


Abdominal Ultrasound

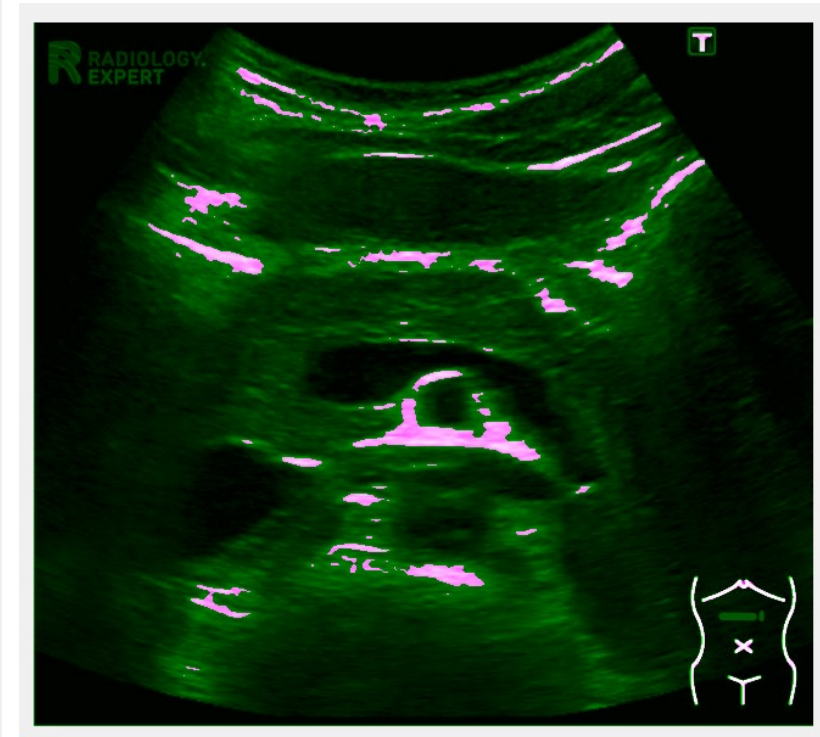
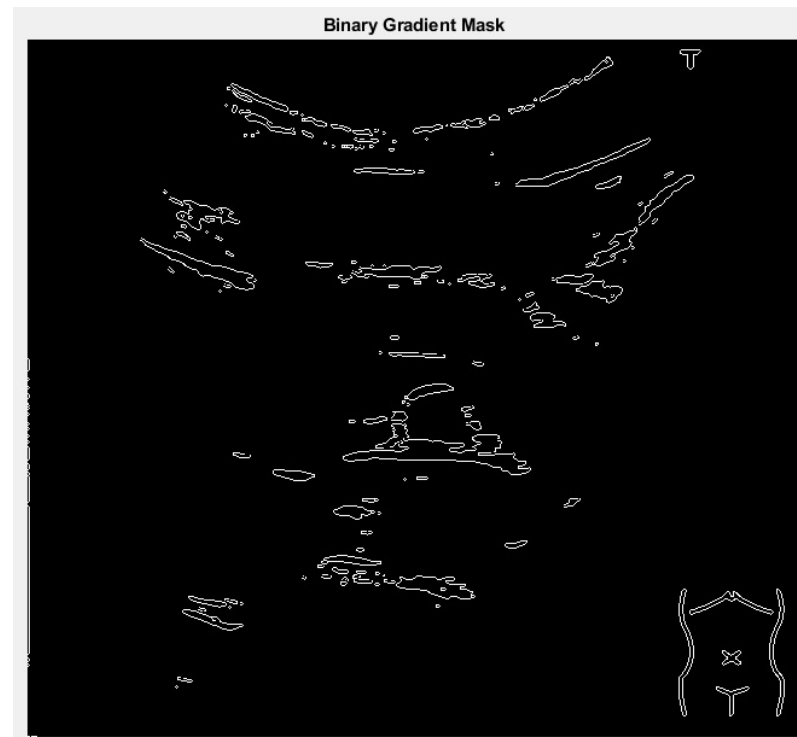
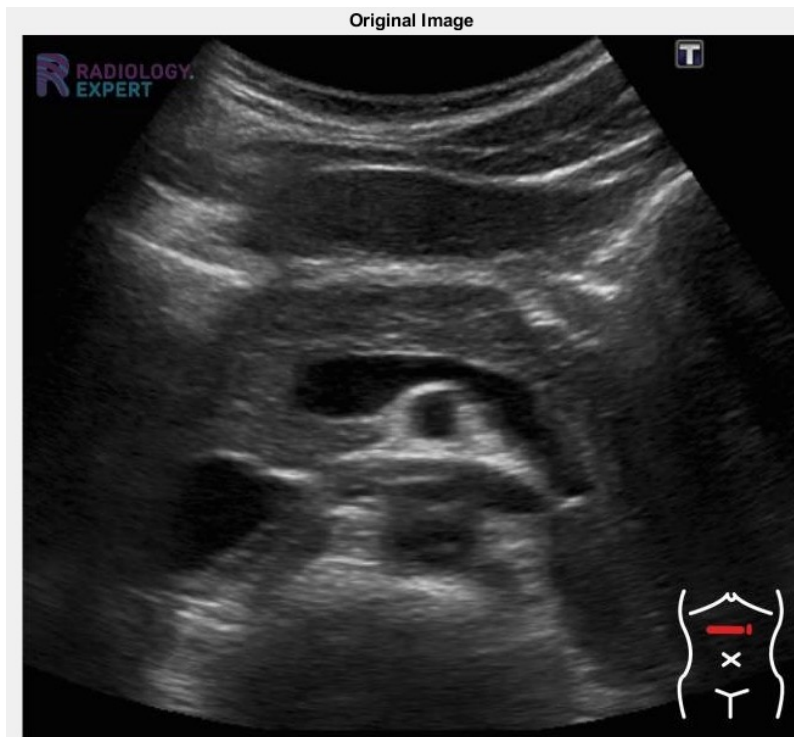
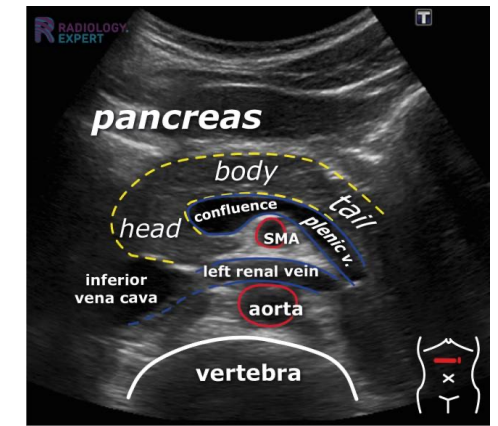




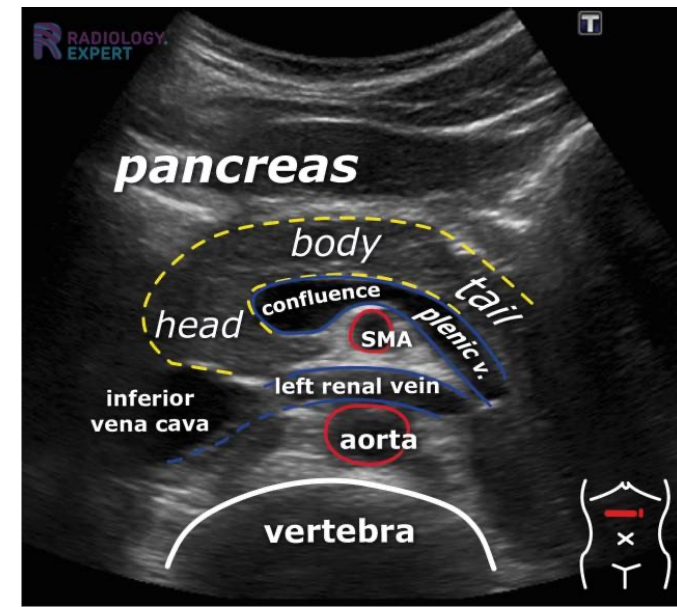
Quantization



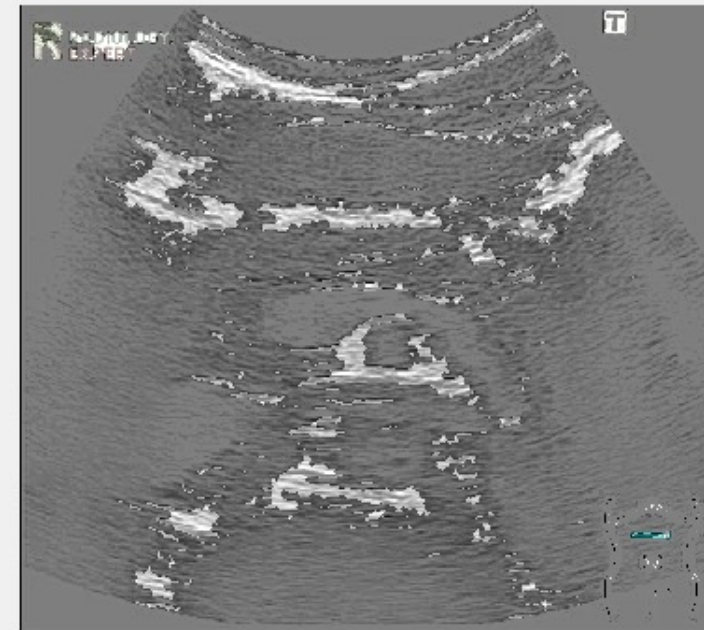
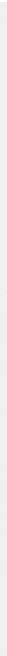
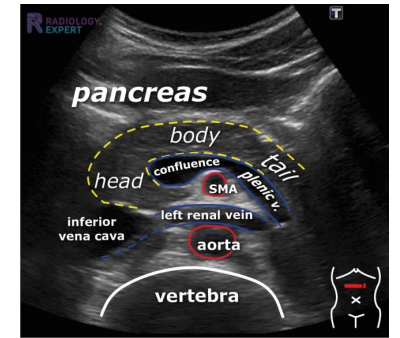
Dilation → Segmentation



Complement → Sharpen



Complement → Sharpen → Dilation





“

Good luck with weekly assignment :)